

Quiz 3: Conditional Expectation, Entropy, Stochastic Processes.

Solutions

Homework for lectures 5 and 6. Notation as in the lectures. Problems are numbered A1–A21, B1–B22, C1–C22; within each group they get harder, and the last one or two of a group need the concept of that group.

Part A: Conditional Expectation

Notation as in the Conditional Expectation lecture. Two dice are Y (first), Z (second), $X = Y + Z$: the two-dice table, rows the events $\{Y = y\}$. Here also $M = \max(Y, Z)$ and $D = Y - Z$.

Conditional expectation given a value

Lecture battery 1

- A1. Two dice. Which cells form the event $\{M = m\}$, and how many are there? Compute $\mathbb{E}[Z \mid M = m]$ for $m = 1, \dots, 6$ (a formula in m), then $\mathbb{E}[X \mid M = m]$ by symmetry, and deduce $\mathbb{E}[\min(Y, Z) \mid M = m]$ without further counting.

Solution. $\{M = m\}$ is the L -shape $\{(m, z) : z < m\} \cup \{(y, m) : y < m\} \cup \{(m, m)\}$, $2m - 1$ cells, each of conditional probability $\frac{1}{2m-1}$. On it Z takes the values $1, \dots, m - 1$ once each and the value m on m cells, so

$$\mathbb{E}[Z \mid M = m] = \frac{\frac{1}{2}m(m-1) + m^2}{2m-1} = \frac{m(3m-1)}{2(2m-1)} : \quad 1, \frac{5}{3}, \frac{12}{5}, \frac{22}{7}, \frac{35}{9}, \frac{51}{11}.$$

The L -shape is symmetric in Y and Z , so $\mathbb{E}[Y \mid M = m] = \mathbb{E}[Z \mid M = m]$ and $\mathbb{E}[X \mid M = m] = \frac{m(3m-1)}{2m-1}$. Since $\min(Y, Z) = X - M$ and M is the constant m on the row, $\mathbb{E}[\min(Y, Z) \mid M = m] = \frac{m(3m-1)}{2m-1} - m = \frac{m(m-1)}{2m-1} : 0, \frac{2}{3}, \frac{6}{5}, \frac{12}{7}, \frac{20}{9}, \frac{30}{11}$.

- A2. Let Y be uniform on $[1, 2]$ and, given $Y = y$, let X be uniform on $[0, y]$. Sketch the region where $f_{X,Y} > 0$ and write down $f_{X,Y}$ and $f_{X|Y}$ with their domains. Compute $\mathbb{E}[X \mid Y = y]$ and $\mathbb{E}[X^2 \mid Y = y]$.

Solution. $f_Y = 1$ on $[1, 2]$, $f_{X|Y}(x \mid y) = \frac{1}{y}$ for $0 \leq x \leq y$, so $f_{X,Y}(x, y) = \frac{1}{y}$ on the trapezium $\{1 \leq y \leq 2, 0 \leq x \leq y\}$ and 0 elsewhere; $f_{X|Y} : \mathbb{R} \times [1, 2] \rightarrow [0, \infty)$, x the value of X and y the given value of Y . The slice at height y is uniform on $[0, y]$: $\mathbb{E}[X \mid Y = y] = \frac{y}{2}$ and $\mathbb{E}[X^2 \mid Y = y] = \int_0^y x^2 \frac{1}{y} dx = \frac{y^2}{3}$.

Law of total expectation

Lecture battery 2

- A3. Roll a die, getting N ; then draw N independent numbers uniform on $[0, 1]$ and let L be the largest of them. Compute $\mathbb{E}[L]$ by conditioning on N . (Given $N = n$, $\mathbb{P}(L \leq t \mid N = n) = t^n$, so L has conditional pdf nt^{n-1} on $[0, 1]$.) Then find F_L head-on and check $\mathbb{E}[L] = \int_0^1 (1 - F_L(t)) dt$.

Solution. $\mathbb{E}[L \mid N = n] = \int_0^1 t \cdot nt^{n-1} dt = \frac{n}{n+1}$, so

$$\mathbb{E}[L] = \frac{1}{6} \left(\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \frac{5}{6} + \frac{6}{7} \right) = \frac{617}{840} \approx 0.7345.$$

Head-on: $F_L(t) = \sum_n \mathbb{P}(N = n) \mathbb{P}(L \leq t \mid N = n) = \frac{1}{6}(t + t^2 + \dots + t^6)$, a degree-6 polynomial, and $\int_0^1 (1 - F_L) = 1 - \frac{1}{6} \sum_{n=1}^6 \frac{1}{n+1} = \frac{617}{840}$: the same six fractions, reached after finding a distribution nobody asked for.

- A4. With Y, X as in A2, compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ by total expectation and deduce $\text{Var}(X)$. Then find f_X head-on (it is given by two different formulas on $[0, 1]$ and on $[1, 2]$) and check $\mathbb{E}[X]$.

Solution. $\mathbb{E}[X] = \int_1^2 \frac{y}{2} dy = \frac{3}{4}$, $\mathbb{E}[X^2] = \int_1^2 \frac{y^2}{3} dy = \frac{7}{9}$, $\text{Var}(X) = \frac{7}{9} - \frac{9}{16} = \frac{31}{144}$. *Head-on:* $f_X(x) = \int f_{X,Y}(x, y) dy$ integrates $\frac{1}{y}$ over those $y \in [1, 2]$ with $y \geq x$: $f_X(x) = \ln 2$ for $0 \leq x \leq 1$ and $f_X(x) = \ln \frac{2}{x}$ for $1 \leq x \leq 2$. Then $\mathbb{E}[X] = \ln 2 \int_0^1 x dx + \int_1^2 x \ln \frac{2}{x} dx = \frac{\ln 2}{2} + \left(\frac{3}{4} - \frac{\ln 2}{2} \right) = \frac{3}{4}$, after an integration by parts that the conditioning route never needed.

- A5. A supplier's defect rate is unknown: model it as Θ uniform on $[0, \frac{1}{5}]$. Given $\Theta = \theta$, the number D of defective items in a sample of 10 is binomial: $\mathbb{E}[D \mid \Theta = \theta] = 10\theta$ and $\mathbb{E}[D^2 \mid \Theta = \theta] = 10\theta(1 - \theta) + 100\theta^2$. Compute $\mathbb{E}[D]$ and $\text{Var}(D)$, and compare $\text{Var}(D)$ with the variance $10 \cdot 0.1 \cdot 0.9$ of a binomial with the average defect rate 0.1. Then compute $\mathbb{P}(D = 0)$ head-on and compare with 0.9^{10} .

Solution. $\mathbb{E}[D] = 10\mathbb{E}[\Theta] = 1$. $\mathbb{E}[D^2] = 10\mathbb{E}[\Theta] + 90\mathbb{E}[\Theta^2] = 1 + 90 \cdot \frac{1}{75} = \frac{11}{5}$ (as $\mathbb{E}[\Theta^2] = \int_0^{1/5} 5\theta^2 d\theta = \frac{1}{75}$), so $\text{Var}(D) = \frac{11}{5} - 1 = \frac{6}{5} = 1.2 > 0.9$: not knowing the rate spreads D out. *Head-on:* $\mathbb{P}(D = 0) = \int_0^{1/5} 5(1 - \theta)^{10} d\theta = \frac{5}{11}(1 - 0.8^{11}) \approx 0.416$, against $0.9^{10} \approx 0.349$; each further value $\mathbb{P}(D = d)$ is another such integral, and $\mathbb{E}[D]$ would be assembled from eleven of them.

Random sums

Lecture battery 5

- A6. A bus carries N passengers, $\mathbb{E}[N] = 25$, the count independent of who the passengers are. Each passenger is an adult (fare £1.50) with probability 0.6 or a child (fare £0.75) with probability 0.4. (a) Find the expected takings. Does the answer change if the passengers' types are dependent on one another (families travel together)? (b) Now suppose that with probability $\frac{1}{4}$ the bus is a school run, in which case $N = 40$ and every passenger is a child; otherwise $N = 20$ and every passenger is an adult. Compute the expected takings by conditioning on the kind of run. Then compute $\mu = \mathbb{E}[X_i]$ for a fixed passenger $i \leq 20$ and $\mathbb{E}[N]$, and explain why $\mu \mathbb{E}[N]$ is not the answer.

Solution. (a) $\mu = 0.6 \cdot 1.5 + 0.4 \cdot 0.75 = \text{£}1.20$, so $\mathbb{E}[S] = \mu \mathbb{E}[N] = \text{£}30$. *Dependence between passengers does not matter: the theorem needs N independent of the fares and equal means, not independence among the X_i .* (b) *Conditioning on the kind of run:* $\mathbb{E}[S] = \frac{1}{4} \cdot 40 \cdot 0.75 + \frac{3}{4} \cdot 20 \cdot 1.5 = 7.5 + 22.5 = \text{£}30$. But $\mathbb{E}[N] = \frac{1}{4} \cdot 40 + \frac{3}{4} \cdot 20 = 25$ and $\mu = \frac{1}{4} \cdot 0.75 + \frac{3}{4} \cdot 1.5 = \text{£}1.3125$, so $\mu \mathbb{E}[N] = \text{£}32.81 \neq 30$. *The count is not independent of the fares: on the rows where N is large the fares are small. The row $\{N = 40\}$ changes the distribution of every X_i (all children), which is exactly what the theorem forbids.*

- A7. Two dice X_1, X_2 are rolled in turn, $\mu = \mathbb{E}[X_i] = 3.5$. In each of the following, S is the sum of the terms that are counted; compute $\mathbb{E}[S]$ by conditioning on the value of the die that decides N , and compare with $\mu \mathbb{E}[N]$. (a) Looking back: $N = 2$ if $X_1 \leq 3$, otherwise

$N = 1$ (the second die is rolled and counted only when the first is small). (b) Looking ahead: $N = 2$ if $X_2 \leq 3$, otherwise $N = 1$ (both dice are rolled; the second is counted only when it is small).

Solution. In both cases $\mathbb{P}(N = 2) = \frac{1}{2}$, $\mathbb{E}[N] = \frac{3}{2}$ and $\mu\mathbb{E}[N] = 5.25$. (a) Condition on $X_1 = k$. If $k \leq 3$: $S = k + X_2$ with X_2 untouched, so $\mathbb{E}[S \mid X_1 = k] = k + 3.5$; if $k \geq 4$: $S = k$. Total expectation: $\mathbb{E}[S] = \frac{1}{6}[(1+2+3) + 3 \cdot 3.5 + (4+5+6)] = \frac{31.5}{6} = 5.25 = \mu\mathbb{E}[N]$. (b) Condition on $X_2 = k$. If $k \leq 3$: $S = X_1 + k$, $\mathbb{E}[S \mid X_2 = k] = 3.5 + k$; if $k \geq 4$: $S = X_1$, $\mathbb{E}[S \mid X_2 = k] = 3.5$. So $\mathbb{E}[S] = 3.5 + \frac{1}{6}(1+2+3) = 4.5 \neq 5.25$. In (b) the row $\{N = 2\} = \{X_2 \leq 3\}$ changes the distribution of the second term (it averages 2 there, not 3.5), so the theorem's assumption fails and its formula is wrong. In (a) the decision uses only what has already been rolled and never a term it is about to count; the formula happens to hold (Wald's identity), but the theorem as stated does not apply to (a) either, since N is not independent of X_1 .

Waiting times

Lecture battery 5

- A8. Let T be the number of rolls of a fair die until two sixes in a row first appear. Show $\mathbb{E}[T] = 42$ by conditioning on the opening (first roll not a six; six then not a six; six, six), as the lecture did for HH.

Solution. Let $m = \mathbb{E}[T]$. The three openings have probabilities $\frac{5}{6}, \frac{5}{36}, \frac{1}{36}$; on the first two the game restarts after 1, respectively 2, rolls, and on the third it is over at roll 2:

$$m = \frac{5}{6}(1 + m) + \frac{5}{36}(2 + m) + \frac{1}{36} \cdot 2 = \frac{35}{36}m + \frac{42}{36}, \quad \text{so } m = 42.$$

- A9. Toss a fair coin until the pattern THH first appears; then the same for THT and for TTT. Compute the three expected waiting times by conditioning on the first toss from each of the three states “no progress”, “T”, “TH” (respectively “TT”). Which pattern arrives soonest, and why are three patterns of the same length not equally quick?

Solution. Write m_0, m_1, m_2 for the expected number of further tosses from the states “no progress”, “T”, and “T then the second letter”. Conditioning on the next toss: THH: $m_0 = 1 + \frac{1}{2}(m_0 + m_1)$, $m_1 = 1 + \frac{1}{2}(m_1 + m_2)$, $m_2 = 1 + \frac{1}{2}(0 + m_1)$, so $(m_0, m_1, m_2) = (8, 6, 4)$. THT: the same first two equations, but $m_2 = 1 + \frac{1}{2}(0 + m_0)$, so $(10, 8, 6)$. TTT: $m_0 = 1 + \frac{1}{2}(m_0 + m_1)$, $m_1 = 1 + \frac{1}{2}(m_0 + m_2)$, $m_2 = 1 + \frac{1}{2}(0 + m_0)$, so $(14, 12, 8)$. So THH takes 8 tosses on average, THT 10, TTT 14. The difference is what a failure leaves behind: after TH, a wrong toss for THH is T, which is the start of a new attempt; a wrong toss for THT is H, which is worth nothing; and for TTT a failing H at any stage throws away all progress.

- A10. Roll a fair die repeatedly. (a) Let T be the number of rolls until both a 1 and a 6 have appeared. Compute $\mathbb{E}[T]$ by conditioning on the first roll. (b) Let T_6 be the number of rolls until every face has appeared at least once. Writing m_k for the expected number of further rolls when k distinct faces have been seen, derive $m_k = \frac{6-k}{6} + m_{k+1}$ and compute $\mathbb{E}[T_6]$. Why is $\mathbb{P}(T_6 = n)$ unpleasant to write down?

Solution. (a) The first roll is a 1 or a 6 with probability $\frac{1}{3}$; then the wait for the other specific face is geometric with mean 6. Otherwise nothing has changed: $\mathbb{E}[T] = \frac{1}{3}(1 + 6) + \frac{2}{3}(1 + \mathbb{E}[T])$, so $\mathbb{E}[T] = 9$. (b) With k faces seen, the next roll is new with probability $\frac{6-k}{6}$: $m_k = 1 + \frac{k}{6}m_k + \frac{6-k}{6}m_{k+1}$, i.e. $m_k = \frac{6-k}{6} + m_{k+1}$, with $m_6 = 0$. Hence $\mathbb{E}[T_6] = m_0 =$

$6(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}) = 14.7$. $\mathbb{P}(T_6 = n)$ is the probability that $n-1$ rolls show exactly five faces and roll n the sixth; counting sequences of $n-1$ rolls that use exactly five given faces needs inclusion-exclusion over the missing faces, for every n , and then $\sum_n n\mathbb{P}(T_6 = n)$ still has to be summed.

Conditional expectation as a random variable

Lecture battery 3

- A11. Two dice, $M = \max(Y, Z)$, and $\mathbb{E}[X | M]$ from A1. Write down $\mathbb{E}[X | M](\omega)$ and $X(\omega)$ for $\omega = (2, 5), (5, 5), (5, 2), (6, 1)$. On which set of outcomes does $\mathbb{E}[X | M]$ take the value $\frac{70}{9}$? Describe the shape of the “rows” that conditioning on M paints, and check that they partition the table.

Solution. $\mathbb{E}[X | M] = \frac{M(3M-1)}{2M-1}$. At $(2, 5), (5, 5), (5, 2)$: $M = 5$, value $\frac{70}{9} \approx 7.78$, while $X = 7, 10, 7$. At $(6, 1)$: $M = 6$, value $\frac{102}{11} \approx 9.27$, while $X = 7$. The value $\frac{70}{9}$ is taken on $\{M = 5\}$: the nine cells $(5, 1), \dots, (5, 5), (1, 5), \dots, (4, 5)$, an L-shape. The rows of this conditioning are the six nested L-shapes $\{M = m\}$, of sizes 1, 3, 5, 7, 9, 11; they are disjoint and their sizes add to 36.

- A12. Two dice. Compute $\mathbb{E}[M | X = x]$ and $\mathbb{E}[\min(Y, Z) | X = x]$ for $x = 4, 7, 10$, by listing the cells of each diagonal. Verify $\mathbb{E}[M | X] + \mathbb{E}[\min(Y, Z) | X] = X$ and explain from the properties of conditional expectation why this holds without any listing. Then write $\mathbb{E}[|D| | X]$ in terms of $\mathbb{E}[M | X]$ and X .

Solution. Diagonal $x = 7$: the cells $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ have $M = 6, 5, 4, 4, 5, 6$ and $\min = 1, 2, 3, 3, 2, 1$, so $\mathbb{E}[M | X = 7] = 5$ and $\mathbb{E}[\min | X = 7] = 2$. Diagonal $x = 4$: the cells $(1, 3), (2, 2), (3, 1)$ give $\mathbb{E}[M | X = 4] = \frac{8}{3}$ and $\mathbb{E}[\min | X = 4] = \frac{4}{3}$. Diagonal $x = 10$: the cells $(4, 6), (5, 5), (6, 4)$ give $\frac{17}{3}$ and $\frac{13}{3}$. In each case the two add to x : $M + \min(Y, Z) = Y + Z = X$, so by linearity $\mathbb{E}[M | X] + \mathbb{E}[\min | X] = \mathbb{E}[X | X] = X$, as X is constant on each of its own rows. Since $|D| = M - \min(Y, Z) = 2M - X$, $\mathbb{E}[|D| | X] = 2\mathbb{E}[M | X] - X$: $\frac{4}{3}, 3, \frac{4}{3}$ at $x = 4, 7, 10$.

- A13. Let U be uniform on $\{1, 2, 3, 4\}$, $V = \mathbf{1}_{\{U \text{ odd}\}}$ and $W = \mathbf{1}_{\{U \geq 3\}}$. List the values of the four random variables $\mathbb{E}[U | V]$, $\mathbb{E}[U | W]$, $\mathbb{E}[V | W]$, $\mathbb{E}[W | V]$ at each of the four outcomes, saying which rows each conditioning creates. Two of the four are constant: what does that suggest about V and W , and is it true?

Solution. V creates the rows $\{1, 3\}$ and $\{2, 4\}$; W the rows $\{1, 2\}$ and $\{3, 4\}$. $\mathbb{E}[U | V]$: 2 on $\{1, 3\}$, 3 on $\{2, 4\}$: values 2, 3, 2, 3 at $U = 1, 2, 3, 4$. $\mathbb{E}[U | W]$: $\frac{3}{2}$ on $\{1, 2\}$, $\frac{7}{2}$ on $\{3, 4\}$: $\frac{3}{2}, \frac{3}{2}, \frac{7}{2}, \frac{7}{2}$. $\mathbb{E}[V | W]$: on $\{1, 2\}$ the values of V are 1, 0, average $\frac{1}{2}$; on $\{3, 4\}$ also $\frac{1}{2}$: constant $\frac{1}{2} = \mathbb{E}[V]$. $\mathbb{E}[W | V]$: on $\{1, 3\}$, $W = 0, 1$; on $\{2, 4\}$, $W = 0, 1$: constant $\frac{1}{2} = \mathbb{E}[W]$. Constancy suggests $V \perp\!\!\!\perp W$, and here it is true: $\mathbb{P}(V = v, W = w) = \frac{1}{4} = \mathbb{P}(V = v)\mathbb{P}(W = w)$ for all four pairs. (A19 shows that constancy alone does not guarantee this.)

The tower property

Lecture battery 4

- A14. Let N be Poisson with mean λ , that is, $\mathbb{P}(N = n) = e^{-\lambda}\lambda^n/n!$ for $n = 0, 1, 2, \dots$, so that $\mathbb{E}[N] = \lambda$. Toss a fair coin N times; H is the number of heads. Compute $\mathbb{E}[H]$. Then show that H is Poisson with mean $\lambda/2$, and compare the effort.

Solution. $\mathbb{E}[H | N] = N/2$, so $\mathbb{E}[H] = \lambda/2$ by the tower property. Head-on: $\mathbb{P}(H = h) = \sum_{n \geq h} e^{-\lambda} \frac{\lambda^n}{n!} \binom{n}{h} 2^{-n} = e^{-\lambda} \frac{(\lambda/2)^h}{h!} \sum_{m \geq 0} \frac{(\lambda/2)^m}{m!} = e^{-\lambda/2} \frac{(\lambda/2)^h}{h!}$, which is Poisson($\lambda/2$), whose

mean is $\lambda/2$. Correct, but several lines and a trick with the exponential series, versus one line.

- A15. A stick of length 1 is broken at a uniform point Y ; the longer piece is kept. Let L be its length. Compute $\mathbb{E}[L]$ two ways: directly from the definition of L , and by conditioning on the event $\{Y < \frac{1}{2}\}$. Then: the kept piece is broken again at a uniform point, and the longer part kept, with length L_2 . Compute $\mathbb{E}[L_2]$.

Solution. $L = \max(Y, 1-Y)$, so $\mathbb{E}[L] = \int_0^{1/2} (1-y) dy + \int_{1/2}^1 y dy = \frac{3}{4}$. Conditioning: given $Y < \frac{1}{2}$, $L = 1 - Y$ is uniform on $(\frac{1}{2}, 1)$ with mean $\frac{3}{4}$; likewise given $Y \geq \frac{1}{2}$; so $\mathbb{E}[L] = \frac{3}{4}$. For the second break, given $L = \ell$ the same argument scaled by ℓ gives $\mathbb{E}[L_2 | L = \ell] = \frac{3}{4}\ell$, so $\mathbb{E}[L_2 | L] = \frac{3}{4}L$ and $\mathbb{E}[L_2] = \frac{3}{4}\mathbb{E}[L] = \frac{9}{16}$ by the tower property.

Taking out what is known

Lecture battery 4

- A16. Two dice, $M = \max(Y, Z)$. Compute $\mathbb{E}[MZ]$ using $\mathbb{E}[h(M)Z] = \mathbb{E}[h(M)\mathbb{E}[Z | M]]$ with $\mathbb{E}[Z | M]$ from A1 and the pmf $p_M(m) = \frac{2m-1}{36}$. Deduce $\text{Cov}(M, Z)$. Check $\mathbb{E}[MZ]$ against the table.

Solution. $\mathbb{E}[MZ] = \mathbb{E}[M\mathbb{E}[Z | M]] = \sum_{m=1}^6 m \cdot \frac{m(3m-1)}{2(2m-1)} \cdot \frac{2m-1}{36} = \frac{1}{72} \sum_{m=1}^6 m^2(3m-1) = \frac{3 \cdot 441 - 91}{72} = \frac{154}{9} \approx 17.11$. $\mathbb{E}[M] = \sum_m m \frac{2m-1}{36} = \frac{161}{36}$, so $\text{Cov}(M, Z) = \frac{154}{9} - \frac{161}{36} \cdot \frac{7}{2} = \frac{35}{24} \approx 1.46$. Check: $\frac{1}{36} \sum_{y,z} \max(y, z) z = \frac{616}{36} = \frac{154}{9}$.

- A17. With Y, X as in A2 (Y uniform on $[1, 2]$, X uniform on $[0, Y]$ given Y), compute $\mathbb{E}[XY]$ and $\mathbb{E}[XY^2]$ by taking out what is known, and deduce $\text{Cov}(X, Y)$. Which double integrals did you avoid?

Solution. $\mathbb{E}[XY] = \mathbb{E}[Y\mathbb{E}[X | Y]] = \mathbb{E}[Y^2/2] = \frac{1}{2} \int_1^2 y^2 dy = \frac{7}{6}$, and $\mathbb{E}[XY^2] = \mathbb{E}[Y^3/2] = \frac{1}{2} \cdot \frac{15}{4} = \frac{15}{8}$. $\text{Cov}(X, Y) = \frac{7}{6} - \frac{3}{4} \cdot \frac{3}{2} = \frac{1}{24}$. Avoided: $\int_1^2 \int_0^y xy \frac{1}{y} dx dy$ and $\int_1^2 \int_0^y xy^2 \frac{1}{y} dx dy$ over the trapezium.

Independence

Lecture battery 4

- A18. Two dice, $D = Y - Z$. Compute $\mathbb{E}[D | Y]$, $\mathbb{E}[D | Z]$ and $\mathbb{E}[D | X]$ as random variables. Which of the three conditionings carries information about D , and which property of conditional expectation settles each one? (For $\mathbb{E}[D | X]$, look at how the diagonal $\{X = x\}$ is placed relative to the main diagonal.)

Solution. $\mathbb{E}[D | Y] = Y - \mathbb{E}[Z | Y] = Y - 3.5$, because Y comes out as known and $Z \perp\!\!\!\perp Y$ gives $\mathbb{E}[Z | Y] = \mathbb{E}[Z]$. Likewise $\mathbb{E}[D | Z] = 3.5 - Z$. On the diagonal $\{X = x\}$ the cells $(y, x-y)$ and $(x-y, y)$ both lie on it and have opposite values of D , so the row average of D is 0: $\mathbb{E}[D | X] = 0$, a constant. The first two conditionings know something about D (each fixes one of its two terms); the third knows nothing about its sign.

- A19. Two dice, $D = Y - Z$, $X = Y + Z$. Show $\mathbb{E}[X | D] = 7 = \mathbb{E}[X]$ (which cells form $\{D = d\}$, and what is X on them?), and recall $\mathbb{E}[D | X] = 0 = \mathbb{E}[D]$ from A18. Deduce $\text{Cov}(X, D) = 0$. Then show that X and D are *not* independent, by comparing the parities of X and D or by computing $\mathbb{P}(D = 5, X = 7)$ against $\mathbb{P}(D = 5)\mathbb{P}(X = 7)$. Which lesson of the lecture is this an instance of?

Solution. $\{D = d\}$ is the diagonal of cells $(y, y-d)$ with both coordinates in $\{1, \dots, 6\}$; on it $X = 2y - d$ and y runs over an interval symmetric about $\frac{7+d}{2}$, so the average of X

is $2 \cdot \frac{7+d}{2} - d = 7$. Hence $\mathbb{E}[X | D] = 7$, and $\text{Cov}(X, D) = \mathbb{E}[D \mathbb{E}[X | D]] - \mathbb{E}[X]\mathbb{E}[D] = 7\mathbb{E}[D] - 7\mathbb{E}[D] = 0$. Not independent: $X = D + 2Z$, so X and D always have the same parity, and $\mathbb{P}(D \text{ odd}, X \text{ even}) = 0 \neq \mathbb{P}(D \text{ odd})\mathbb{P}(X \text{ even})$. Or: $\mathbb{P}(D = 5, X = 7) = \mathbb{P}(Y = 6, Z = 1) = \frac{1}{36}$, while $\mathbb{P}(D = 5)\mathbb{P}(X = 7) = \frac{1}{36} \cdot \frac{6}{36}$. This is “equal row averages are not equal row distributions”: $\mathbb{E}[X | D] = \mathbb{E}[X]$ says every diagonal has average 7, independence would say every diagonal has the distribution of X , and a diagonal with 3 cells cannot.

Putting it together

Lecture battery 5

- A20. A gambler bets on a fair coin and doubles her stake after each loss (stakes $\pounds 1, 2, 4, \dots$), stopping after the first win; R is the number of rounds. (a) Show that her net gain is exactly $\pounds 1$ on every outcome, and compute the expected total amount staked by conditioning on R . (b) Now the table limit stops her after round 5 whether or not she has won. Compute her net gain on each outcome, then $\mathbb{E}[\text{net gain}]$, $\mathbb{E}[R]$ and the expected total staked. What do you conclude about the doubling strategy?

Solution. (a) After k losses she has lost $1 + 2 + \dots + 2^{k-1} = 2^k - 1$ and then wins 2^k : net $+1$, always. The total staked is $2^R - 1$ with $\mathbb{P}(R = r) = 2^{-r}$, and $\mathbb{E}[2^R - 1] = \sum_{r \geq 1} (2^r - 1)2^{-r} = \infty$. (b) With the limit, $R = r$ for $r \leq 4$ still has probability 2^{-r} and $R = 5$ has probability $\frac{1}{16}$: a win at round 5 ($\frac{1}{32}$, gain $+1$) or five losses ($\frac{1}{32}$, gain -31). So $\mathbb{E}[\text{gain}] = \frac{31}{32} \cdot 1 + \frac{1}{32} \cdot (-31) = 0$; $\mathbb{E}[R] = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + 5 \cdot \frac{1}{16} = \frac{31}{16}$; $\mathbb{E}[2^R - 1] = \frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + 31 \cdot \frac{1}{16} = 5$. The sure $\pounds 1$ of (a) is bought with unbounded expected exposure; any finite limit restores $\mathbb{E}[\text{gain}] = 0$ exactly, by trading a $\frac{1}{32}$ chance of losing $\pounds 31$ for the $\frac{31}{32}$ chance of winning $\pounds 1$. No stake-sizing rule makes a fair game favourable.

- A21. Let $X_1, X_2, \dots$ be independent, each uniform on $\{1, \dots, 6\}$, and let N be the index of the first roll showing 5 or 6; N is determined by the X_i and not independent of them. (a) Let $S = X_1 + \dots + X_N$ include the stopping roll. Compute $\mathbb{E}[S]$ by conditioning on the first roll, and compare with $\mu \mathbb{E}[N]$. (b) Let $S' = X_1 + \dots + X_{N-1}$ exclude it. Compute $\mathbb{E}[S']$ the same way and compare with $\mu \mathbb{E}[N - 1]$. Explain the two comparisons.

Solution. N is geometric with success probability $\frac{1}{3}$: $\mathbb{E}[N] = 3$, $\mu \mathbb{E}[N] = 10.5$, $\mu \mathbb{E}[N - 1] = 7$. (a) Condition on $X_1 = k$. If $k \in \{5, 6\}$: $S = k$. If $k \leq 4$: $S = k + S''$ with S'' a fresh copy of S . So $\mathbb{E}[S] = \frac{1}{6}(5 + 6) + \frac{1}{6} \sum_{k=1}^4 (k + \mathbb{E}[S]) = \frac{11}{6} + \frac{10}{6} + \frac{4}{6} \mathbb{E}[S]$, giving $\mathbb{E}[S] = \frac{21}{2} = 10.5 = \mu \mathbb{E}[N]$. (b) Now $S' = 0$ if $k \in \{5, 6\}$ and $S' = k + S'''$ otherwise: $\mathbb{E}[S'] = \frac{10}{6} + \frac{4}{6} \mathbb{E}[S']$, so $\mathbb{E}[S'] = 5 \neq 7$. The random-sums theorem applies to neither sum, since N depends on the X_i . In (a) the stopping rule never looks at a roll before deciding to count it, and the formula still holds (Wald's identity). In (b) the last roll is dropped because it was large: $\mathbb{E}[S'] = \mathbb{E}[S] - \mathbb{E}[X_N] = 10.5 - 5.5 = 5$, and the dropped term has mean 5.5, not μ . The formula counts $N - 1$ typical terms; the actual terms are the small ones.

Part B: Entropy

Logarithms are base 2, the unit is the bit, and $0 \log 0 = 0$. Notation as in the Entropy lecture: p_X is the pmf of X and S_X its support; $I_X(x) = \log \frac{1}{p_X(x)}$; $H(X)$ is entropy and $h(p)$ the binary entropy function; $H(P, Q)$ is the cross-entropy of the truth P with the code or model Q , $D(P \| Q) = H(P, Q) - H(P)$; $H(X | Y)$ and $I(X; Y)$ are conditional entropy and mutual information. Two dice are always Y (first), Z (second), $X = Y + Z$, the two-dice table.

- B1. A source emits one of five symbols a, b, c, d, e with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}$. (a) Check that the lengths $(1, 2, 3, 4, 4)$ satisfy Kraft's inequality with equality and write down a prefix-free code c with these lengths. (b) Compute $L(c)$ and $H(X)$. (c) Decode 11100101111100 with your code. (d) Compute $\text{Var}(\ell_c(X))$ and the mean and standard deviation of the number of bits needed for 1000 symbols.

Solution. (a) $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} = 1$. Code: $a \mapsto 0, b \mapsto 10, c \mapsto 110, d \mapsto 1110, e \mapsto 1111$; no codeword begins another. (b) $L(c) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{16} \cdot 4 + \frac{1}{16} \cdot 4 = \frac{15}{8} = 1.875$ bits, and $H(X) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + 2 \cdot \frac{1}{16} \cdot 4 = 1.875$: the lengths are the informations, so $L = H$ and the code is optimal. (c) Read left to right, stopping at each leaf: 1110|0|10|1111|110|0: $dabeca$. (d) $\mathbb{E}[\ell_c(X)^2] = \frac{1}{2} + \frac{4}{4} + \frac{9}{8} + \frac{16}{16} + \frac{16}{16} = 4.625$, so $\text{Var} = 4.625 - 1.875^2 = 1.109$. For 1000 independent symbols: mean 1875 bits, standard deviation $\sqrt{1000 \cdot 1.109} = 33.3$ bits. A fixed-length code needs 3000 bits.

- B2. (a) Which of the following can be the lengths of a prefix-free code: $(1, 2, 3, 3, 4)$ for five symbols; $(2, 2, 2, 3, 3, 3)$ for six; $(2, 2, 3, 3, 3, 3)$ for six; $(1, 3, 3, 3, 3)$ for five; $(1, 2, 3, 4, 5, 5)$ for six? For each admissible profile write down a code. (b) Show that if a prefix-free code has $\sum_i 2^{-\ell_i} < 1$, then there is a bit string that is neither a prefix of a codeword nor has a codeword as a prefix, so a further symbol could be added to the code without lengthening any existing codeword.

Solution. (a) Kraft sums: $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} = \frac{17}{16} > 1$, impossible. $3 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} = \frac{9}{8} > 1$, impossible. $2 \cdot \frac{1}{4} + 4 \cdot \frac{1}{8} = 1$: 00, 01, 100, 101, 110, 111. $\frac{1}{2} + 4 \cdot \frac{1}{8} = 1$: 0, 100, 101, 110, 111. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{2}{32} = 1$: 0, 10, 110, 1110, 11110, 11111. (b) Let $L = \max_i \ell_i$. Of the 2^L strings of length L , exactly $2^{L-\ell_i}$ have codeword i as a prefix, and these sets are disjoint (prefix-free), so they cover $2^L \sum_i 2^{-\ell_i} < 2^L$ strings. Any uncovered string w of length L has no codeword as a prefix; and since $|w| = L \geq \ell_i$, w is a prefix of a codeword only if it equals one, which it does not. Adding w as a codeword keeps the code prefix-free.

- B3. Optimality of the rover's code. Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. (a) Using Kraft's inequality and Gibbs' lemma, show that every prefix-free code for four symbols has $L(c) \geq 1.75$ under p , so 0, 10, 110, 111 is optimal. (b) Show that the only length profiles for four symbols with Kraft sum exactly 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$, up to order. (Consider the shortest length.) (c) Take as given that an optimal code has Kraft sum 1 (otherwise a codeword can be shortened). Deduce that for every pmf on four symbols the optimal code is either the fixed-length code or 0, 10, 110, 111 with the symbols assigned in decreasing order of probability, whichever has the smaller L . (d) Apply (c) to $p = (0.45, 0.25, 0.15, 0.15)$ and to $p = (0.3, 0.3, 0.2, 0.2)$, and compare L^* with H in each case. Why can the source coding theorem alone not decide which code is optimal here?

Solution. (a) Put $a(x) = 2^{-\ell_c(x)}$; Kraft gives $\sum a \leq 1$ and $\ell_c(x) = \log \frac{1}{a(x)}$. Gibbs' lemma: $L(c) = \sum_x p(x) \log \frac{1}{a(x)} \geq \sum_x p(x) \log \frac{1}{p(x)} = H(p) = 1.75$, with equality iff $a = p$, i.e. lengths $(1, 2, 3, 3)$. (b) Order $\ell_1 \leq \ell_2 \leq \ell_3 \leq \ell_4$. If $\ell_1 \geq 2$ then $\sum 2^{-\ell_i} \leq 4 \cdot \frac{1}{4} = 1$ with equality only for $(2, 2, 2, 2)$. If $\ell_1 = 1$, the other three sum to $\frac{1}{2}$; $\ell_2 = 1$ is impossible (two words of length 1 already fill the Kraft sum and leave nothing for ℓ_3, ℓ_4), and if $\ell_2 \geq 3$ they sum to at most $\frac{3}{8}$, so $\ell_2 = 2$ and $2^{-\ell_3} + 2^{-\ell_4} = \frac{1}{4}$ with $\ell_3, \ell_4 \geq 3$, forcing $(3, 3)$. Hence $(1, 2, 3, 3)$. (c) An optimal code has one of the two profiles of (b). For $(1, 2, 3, 3)$, $L = \ell_1 p_{(1)} + \ell_2 p_{(2)} + 3p_{(3)} + 3p_{(4)}$ is smallest when the shortest words go to the most probable symbols (swapping two assignments that violate this can only lower L). So $L^* = \min(2, p_{(1)} + 2p_{(2)} + 3p_{(3)} + 3p_{(4)})$ with $p_{(1)} \geq \dots \geq p_{(4)}$. (d) $(0.45, 0.25, 0.15, 0.15)$: $L(1, 2, 3, 3) = 0.45 + 0.5 + 0.45 + 0.45 = 1.85 < 2$, so 0, 10, 110, 111 is optimal, $L^* = 1.85$,

while $H = 1.840$. $(0.3, 0.3, 0.2, 0.2)$: $L(1, 2, 3, 3) = 0.3 + 0.6 + 0.6 + 0.6 = 2.1 > 2$, so the fixed-length code is optimal, $L^* = 2$, $H = 1.971$. The theorem only says $H \leq L^* < H + 1$, i.e. $L^* \in [1.84, 2.84]$ and $[1.97, 2.97]$: both candidates lie in the window. Deciding needs the enumeration of full trees, not the entropy bound.

Information

Lecture Section 2

- B4. Let $f: (0, 1] \rightarrow [0, \infty)$ be non-increasing with $f(pq) = f(p) + f(q)$ and $f(\frac{1}{2}) = 1$. (a) Compute $f(\frac{1}{4})$, $f(\frac{1}{64})$ and $f(2^{-1/3})$. (b) Using $2^7 < 3^5 < 2^8$, show $\frac{7}{5} \leq f(\frac{1}{3}) \leq \frac{8}{5}$. (c) Using $2^{19} < 3^{12} < 2^{20}$, sharpen this to $\frac{19}{12} \leq f(\frac{1}{3}) \leq \frac{20}{12}$, and compare with $\log 3 = 1.585$. (d) $g(p) = \frac{1}{p} - 1$ is decreasing with $g(\frac{1}{2}) = 1$. Which requirement does it fail, and at which numbers?

Solution. (a) $f(\frac{1}{4}) = 2f(\frac{1}{2}) = 2$; $f(\frac{1}{64}) = 6$; $3f(2^{-1/3}) = f(\frac{1}{2}) = 1$, so $f(2^{-1/3}) = \frac{1}{3}$. (b) $f(3^{-5}) = 5f(\frac{1}{3})$. Since $2^{-8} < 3^{-5} < 2^{-7}$ and f is non-increasing, $f(2^{-7}) \leq f(3^{-5}) \leq f(2^{-8})$, i.e. $7 \leq 5f(\frac{1}{3}) \leq 8$. (c) Likewise $2^{-20} < 3^{-12} < 2^{-19}$ gives $19 \leq 12f(\frac{1}{3}) \leq 20$, so $f(\frac{1}{3}) \in [1.583, 1.667]$; indeed $\log 3 = 1.585$. (d) Additivity: $g(\frac{1}{4}) = 3 \neq 2 = g(\frac{1}{2}) + g(\frac{1}{2})$. Two independent fair coin tosses would carry 3 units of information, not 2.

- B5. Two dice. Let $A = \{\text{at least one six}\}$, $B = \{Y = Z\}$, $C = \{Y \text{ even}\}$, $D = \{X \text{ odd}\}$, $E = \{X \geq 10\}$, $F = \{Y = 6\}$. (a) Compute $I(A)$, $I(B)$ and $I(A \cap B)$. Do the informations add? Why not? (b) Compute $I(C)$, $I(D)$ and $I(C \cap D)$, and explain the result by listing the cells of $C \cap D$. (c) Compute $I(E)$, $I(F)$ and $I(E \cap F)$. Check that $I(E \cap F) = I(F) + \log \frac{1}{\mathbb{P}(E|F)}$ and say in words why $E \cap F$ is less surprising than $I(E) + I(F)$ would suggest.

Solution. (a) $\mathbb{P}(A) = \frac{11}{36}$, $I(A) = 1.710$; $\mathbb{P}(B) = \frac{6}{36}$, $I(B) = 2.585$; $A \cap B = \{(6, 6)\}$, $I(A \cap B) = \log 36 = 5.170 \neq 1.710 + 2.585 = 4.295$. They do not add because A, B are not independent: $\mathbb{P}(A \cap B) = \frac{1}{36} \neq \frac{11}{36} \cdot \frac{1}{6}$. (b) $I(C) = I(D) = 1$ bit. X is odd iff Y and Z have different parities, so $C \cap D = \{Y \text{ even}, Z \text{ odd}\}$: $3 \times 3 = 9$ cells, probability $\frac{1}{4}$, $I(C \cap D) = 2 = 1 + 1$. The events are independent. (c) E has the six cells $(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)$: $I(E) = \log 6 = 2.585 = I(F)$. $E \cap F = \{(6, 4), (6, 5), (6, 6)\}$: $I(E \cap F) = \log 12 = 3.585$, against 5.170. Indeed $\mathbb{P}(E | F) = \frac{3}{6} = \frac{1}{2}$ and $2.585 + 1 = 3.585$. Once the first die is a six, a sum of at least 10 is an even chance, so learning E on top of F adds only one bit.

- B6. The rover's source, $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$, coded with 0, 10, 110, 111. Let $L = \ell_c(X)$ be the codeword length of one instruction. (a) Write down the pmf of L and compute $\mathbb{E}[L]$ and $\text{Var}(L)$. (b) For 1000 instructions, give the mean and standard deviation of the total number of bits, and compare with the fixed-length code. (c) Show that a stream of independent fair bits cannot be compressed: any prefix-free code for blocks of k bits has average length at least k . Why does it follow that compressing the rover's coded stream a second time gains nothing?

Solution. (a) $\mathbb{P}(L = 1) = \frac{1}{2}$, $\mathbb{P}(L = 2) = \frac{1}{4}$, $\mathbb{P}(L = 3) = \frac{1}{4}$. $\mathbb{E}[L] = 1.75 = H(X)$; $\mathbb{E}[L^2] = \frac{1}{2} + 1 + \frac{9}{4} = 3.75$, $\text{Var}(L) = 3.75 - 1.75^2 = 0.6875$. (b) Mean 1750 bits, standard deviation $\sqrt{1000 \cdot 0.6875} = 26.2$ bits, so the total is within about 1700–1800 with high probability, against 2000 for the fixed-length code. (c) A block of k independent fair bits is uniform on 2^k values, so its entropy is k ; the source coding lower bound gives $L(c) \geq k$ for every prefix-free code of blocks. The lecture showed that every bit of the rover's coded stream is a fair coin flip independent of the earlier bits, so the coded stream is exactly such a source: no code for it averages under one bit per bit.

- B7. Compute the following entropies. (a) $H(0.4, 0.3, 0.2, 0.1)$. (b) N = the number of heads in three tosses of a fair coin. Compute $H(N)$ and compare with the entropy of the three tosses themselves. Where did the missing bits go? (Compute $H(\text{tosses} \mid N)$.) (c) $M = \min(Y, Z)$ for two dice. Compute $H(M)$ from its pmf and compare it with $H(Y)$ and with $H(\max(Y, Z))$.

Solution. (a) $0.4 \log 2.5 + 0.3 \log \frac{10}{3} + 0.2 \log 5 + 0.1 \log 10 = 0.529 + 0.521 + 0.464 + 0.332 = 1.846$ bits. (b) $p_N = (\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$: $H(N) = 2 \cdot \frac{1}{8} \cdot 3 + 2 \cdot \frac{3}{8} \log \frac{8}{3} = 0.75 + 0.75 \cdot 1.415 = 1.811$ bits. The tosses have $H = 3$ bits. Given $N = n$ the tosses are uniform on the $\binom{3}{n}$ orderings, so $H(\text{tosses} \mid N) = 2 \cdot \frac{3}{8} \log 3 = 1.189$; by the chain rule $3 = H(N) + 1.189$: the missing bits are the order of the heads, which the count forgets. (c) $\mathbb{P}(M = m) = \frac{13-2m}{36}$: pmf $(11, 9, 7, 5, 3, 1)/36$ and $H(M) = 2.320$ bits, less than $H(Y) = \log 6 = 2.585$ (a function of the pair that is far from uniform). $\max(Y, Z)$ has the mirror pmf $(1, 3, 5, 7, 9, 11)/36$, hence the same entropy 2.320.

- B8. Coin A shows heads with probability 0.9, coin B is fair. One coin is chosen uniformly at random ($C \in \{A, B\}$) and tossed once, giving $X \in \{H, T\}$. (a) Compute the pmf of X and $H(X)$. (b) Compute $H(X \mid C = A)$, $H(X \mid C = B)$ and $H(X \mid C)$. (c) Compare $H(X)$ with $H(X \mid C)$ and state the property of h that this illustrates. What is $I(X; C)$? (d) Compute $\mathbb{P}(C = A \mid X = H)$ and $\mathbb{P}(C = A \mid X = T)$, then $H(C \mid X)$, and check $H(C) - H(C \mid X) = I(X; C)$.

Solution. (a) $\mathbb{P}(X = H) = \frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.5 = 0.7$; $H(X) = h(0.7) = 0.881$ bits. (b) $h(0.9) = 0.469$, $h(0.5) = 1$, $H(X \mid C) = \frac{1}{2}(0.469 + 1) = 0.734$. (c) $H(X) = 0.881 > 0.734 = H(X \mid C)$: the entropy of a mixture exceeds the average of the entropies, which is the concavity of h : $h(\frac{0.9+0.5}{2}) \geq \frac{1}{2}h(0.9) + \frac{1}{2}h(0.5)$. The difference $I(X; C) = 0.147$ bits is what one toss tells you about which coin was chosen. (d) $\mathbb{P}(A \mid H) = \frac{0.45}{0.7} = \frac{9}{14} = 0.643$, $\mathbb{P}(A \mid T) = \frac{0.05}{0.3} = \frac{1}{6}$. $H(C \mid X) = 0.7 h(\frac{9}{14}) + 0.3 h(\frac{1}{6}) = 0.7 \cdot 0.940 + 0.3 \cdot 0.650 = 0.853$, and $H(C) - H(C \mid X) = 1 - 0.853 = 0.147 = I(X; C)$.

- B9. Roll a fair die until the first six; N is the number of rolls, so $\mathbb{P}(N = k) = (\frac{5}{6})^{k-1} \frac{1}{6}$ for $k \geq 1$. (a) Write $I_N(k)$ as a function of k . (b) Compute $H(N) = \mathbb{E}[I_N(N)]$ without summing a series, using $\mathbb{E}[N] = 6$. (c) Show that for a geometric variable with success probability p , $H(N) = h(p)/p$, and check the case $p = \frac{1}{2}$. (d) Compute $H(\min(N, 6))$, the entropy of reporting the value only up to “6 or more”, and compare.

Solution. (a) $I_N(k) = \log 6 + (k-1) \log \frac{6}{5}$. (b) $H(N) = \mathbb{E}[\log 6 + (N-1) \log \frac{6}{5}] = \log 6 + (\mathbb{E}[N] - 1) \log \frac{6}{5} = 2.585 + 5 \cdot 0.263 = 3.900$ bits. (c) $I_N(k) = \log \frac{1}{p} + (k-1) \log \frac{1}{q}$ with $q = 1 - p$, and $\mathbb{E}[N] - 1 = \frac{q}{p}$, so $H(N) = \log \frac{1}{p} + \frac{q}{p} \log \frac{1}{q} = \frac{1}{p}(p \log \frac{1}{p} + q \log \frac{1}{q}) = \frac{h(p)}{p}$. For $p = \frac{1}{2}$: $h(\frac{1}{2})/\frac{1}{2} = 2$ bits (rolls until the first head). (d) Pmf $(\frac{1}{6}, \frac{5}{36}, \frac{25}{216}, \frac{125}{1296}, \frac{625}{7776}, (\frac{5}{6})^5)$ with $(\frac{5}{6})^5 = 0.402$: $H = 2.333$ bits. Reporting the exact value costs 1.57 bits more, almost all of it spent on the rare long waits.

The source coding theorem

- B10. Let $p = (0.5, 0.2, 0.2, 0.1)$. (a) Compute $H(p)$. (b) Compute Shannon's lengths $\lceil \log \frac{1}{p(x)} \rceil$, their Kraft sum, a code with these lengths, and its average length. (c) Using B3(c), find the optimal code and L^* . (d) Verify $H \leq L^* \leq L_{\text{Shannon}} < H + 1$. Which of these would be an equality if fractional lengths were allowed?

Solution. (a) $H = 0.5 \cdot 1 + 2 \cdot 0.2 \cdot 2.322 + 0.1 \cdot 3.322 = 1.761$ bits. (b) Ideal lengths 1, 2.32, 2.32, 3.32, rounded up to (1, 3, 3, 4); Kraft sum $\frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} = \frac{13}{16}$; code 0, 100, 101, 1100; $L = 0.5 + 0.6 + 0.6 + 0.4 = 2.1$. (c) Profile (1, 2, 3, 3) in decreasing order of probability: $L = 0.5 \cdot 1 + 0.2 \cdot 2 + 0.2 \cdot 3 + 0.1 \cdot 3 = 1.8 < 2$, so $L^* = 1.8$ with the code 0, 10, 110, 111. (d) $1.761 \leq 1.8 \leq 2.1 < 2.761$. With fractional lengths $\log \frac{1}{p(x)}$ the average length would be exactly H : the first inequality would be an equality.

- B11. A fair die is rolled repeatedly and the results are to be transmitted. Take as given that for M equiprobable symbols the optimal prefix-free code uses only the lengths $\lfloor \log M \rfloor$ and $\lceil \log M \rceil$ and has Kraft sum 1. (a) Single rolls, $M = 6$: find how many codewords have each length and compute L^* per roll. (b) Pairs of rolls, $M = 36$: the same; give the cost per roll. (c) Triples, $M = 216$: the same. (d) Compare with Shannon's construction (all lengths $\lceil \log M \rceil$) for each block size, and with the floor $\log 6 = 2.585$. Which theorem guarantees that the cost per roll tends to 2.585?

Solution. Let $\ell = \lfloor \log M \rfloor$ and let k codewords have length ℓ , the other $M - k$ length $\ell + 1$. Kraft sum 1: $\frac{k}{2^\ell} + \frac{M-k}{2^{\ell+1}} = 1$, so $k = 2^{\ell+1} - M$. (a) $M = 6$, $\ell = 2$: $k = 2$ words of length 2, 4 of length 3; $L^* = \frac{2 \cdot 2 + 4 \cdot 3}{6} = \frac{8}{3} = 2.667$ bits per roll. (b) $M = 36$, $\ell = 5$: $k = 28$ of length 5, 8 of length 6; $L^* = \frac{140 + 48}{36} = \frac{47}{9} = 5.222$ per pair, 2.611 per roll. (c) $M = 216$, $\ell = 7$: $k = 40$ of length 7, 176 of length 8; $L^* = \frac{280 + 1408}{216} = \frac{211}{27} = 7.815$ per triple, 2.605 per roll. (d) Shannon's lengths $\lceil \log M \rceil = 3, 6, 8$ give 3, 3, 2.667 bits per roll: worse than the optimal codes, and for pairs no better than single rolls. The optimal costs 2.667, 2.611, 2.605 descend towards 2.585; the block-coding theorem $H \leq L_k^*/k < H + \frac{1}{k}$ guarantees the limit.

Cross-entropy and relative entropy

Lecture Section 5

- B12. $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ and $Q = (\frac{1}{8}, \frac{1}{8}, \frac{3}{4})$ on $\{a, b, c\}$. (a) Compute $H(P)$ and $H(Q)$. (b) Compute $H(P, Q)$ and $D(P \| Q)$. (c) Compute $H(Q, P)$ and $D(Q \| P)$. (d) Say in words which code is used on which source in (b) and in (c). (e) P 's ideal lengths (1, 2, 2) are integers, Q 's are not. Find the best prefix-free code for Q (B3(b) for three symbols: the only complete profile is (1, 2, 2)), its cost on Q , and its cost on P . Why is the last number smaller than $H(P, Q)$?

Solution. (a) $H(P) = \frac{1}{2} + \frac{2}{4} + \frac{2}{4} = 1.5$. $H(Q) = 2 \cdot \frac{1}{8} \cdot 3 + \frac{3}{4} \log \frac{4}{3} = 0.75 + 0.311 = 1.061$. (b) $H(P, Q) = \frac{1}{2} \cdot 3 + \frac{1}{4} \cdot 3 + \frac{1}{4} \log \frac{4}{3} = 1.5 + 0.75 + 0.104 = 2.354$; $D(P \| Q) = 2.354 - 1.5 = 0.854$. (c) $H(Q, P) = \frac{1}{8} \cdot 1 + \frac{1}{8} \cdot 2 + \frac{3}{4} \cdot 2 = 1.875$; $D(Q \| P) = 1.875 - 1.061 = 0.814$. (d) In (b) the code built for Q (short word for c , three-bit words for a, b) is used on a source that emits a half the time: 0.854 bits per symbol wasted. In (c) the code built for P ($a \mapsto 0$, $b \mapsto 10$, $c \mapsto 11$) is used on a source that emits c three quarters of the time and pays two bits for it each time. (e) Profile (1, 2, 2) with the short word on c : $c \mapsto 0$, $a \mapsto 10$, $b \mapsto 11$. On Q : $\frac{3}{4} \cdot 1 + \frac{1}{8} \cdot 2 + \frac{1}{8} \cdot 2 = 1.25$ bits (against $H(Q) = 1.061$). On P : $\frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 1 = 1.75$ bits, waste 0.25 over $H(P)$. This is less than $D(P \| Q) = 0.854$ because the integer code corresponds to $a' = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$, a less lopsided distribution than Q : rounding the lengths hedged the bet.

- B13. (a) $P = (0.9, 0.05, 0.05)$ and $U = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Compute $H(P)$, $H(P, U)$, $D(P \| U)$, $H(U, P)$ and $D(U \| P)$. The lecture exercises gave $D(P \| U) = \log n - H(P)$ for the uniform U on n symbols; show that the other direction is $D(U \| P) = -\log n - \frac{1}{n} \sum_x \log P(x)$, and say which of the two is unbounded as P moves towards the boundary of Δ_n . (b) Two coins, $A = (0.99, 0.01)$ and $B = (0.9, 0.1)$. Compute $D(A \| B)$ and $D(B \| A)$ and explain the asymmetry in one sentence each way. (c) A model Q is to be scored against the truth P . Which of $D(P \| Q)$, $D(Q \| P)$ measures the bits it wastes, and why must the truth be the first argument?

Solution. (a) $H(P) = 0.9 \log \frac{1}{0.9} + 0.1 \log 20 = 0.137 + 0.432 = 0.569$. $H(P, U) = \log 3 = 1.585$ (every length is $\log 3$), so $D(P\|U) = 1.016$. $H(U, P) = \frac{1}{3}(0.152 + 4.322 + 4.322) = 2.932$, $D(U\|P) = 2.932 - 1.585 = 1.347$. In general $D(U\|P) = \sum_x \frac{1}{n} \log \frac{1/n}{P(x)} = -\log n - \frac{1}{n} \sum_x \log P(x)$; here $-1.585 + \frac{1}{3}(0.152 + 4.322 + 4.322) = 1.347$ again. $D(P\|U) = \log n - H(P) \leq \log n$ is bounded, while $D(U\|P) \rightarrow \infty$ as any $P(x) \rightarrow 0$: coding the uniform source with a code that has given some symbol a very long word is arbitrarily wasteful. (b) $D(A\|B) = 0.99 \log \frac{0.99}{0.9} + 0.01 \log \frac{0.01}{0.1} = 0.136 - 0.033 = 0.103$; $D(B\|A) = 0.9 \log \frac{0.9}{0.99} + 0.1 \log \frac{0.1}{0.01} = -0.124 + 0.332 = 0.208$. Coding the nearly-certain coin A with B 's lengths overpays slightly on the common outcome (0.152 instead of 0.014 bits) almost every time. Coding B with A 's lengths charges $\log 100 = 6.64$ bits for the tail, which comes one time in ten, and that is where the larger waste comes from. (c) $D(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$: the weights must be the frequencies with which symbols actually occur, and those come from P ; Q only supplies the lengths $\log \frac{1}{Q(x)}$ that the model's code would use.

- B14. The rover has two missions, equally likely: $P_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ and $P_2 = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ for (up, down, left, right). One code, built for a pmf Q , must be installed before the mission is known; its cost is $\frac{1}{2}H(P_1, Q) + \frac{1}{2}H(P_2, Q)$ bits per instruction. (a) Show that this cost equals $H(\bar{P}, Q)$ with $\bar{P} = \frac{1}{2}(P_1 + P_2)$, and compute $\bar{P}$. (b) Find the Q that minimises the cost and the minimal cost. Compare with keeping the old code ($Q = P_1$), with the fixed-length code, and with the 1.75 bits achievable if the mission were known. (c) Show that the average waste $\frac{1}{2}D(P_1\|\bar{P}) + \frac{1}{2}D(P_2\|\bar{P})$ equals $I(M; X)$, the mutual information between the mission $M \in \{1, 2\}$ and one instruction X , and compute it. (d) Codeword lengths must be integers. Using B3(b), find the best single-symbol code for the two-mission problem. What must be done to approach the cost in (b)?

Solution. (a) $\frac{1}{2} \sum_x P_1(x) \log \frac{1}{Q(x)} + \frac{1}{2} \sum_x P_2(x) \log \frac{1}{Q(x)} = \sum_x \bar{P}(x) \log \frac{1}{Q(x)}$: cross-entropy is linear in its first argument. $\bar{P} = (\frac{5}{16}, \frac{3}{16}, \frac{3}{16}, \frac{5}{16})$. (b) By Gibbs, $H(\bar{P}, Q) \geq H(\bar{P})$ with equality iff $Q = \bar{P}$. Minimal cost $H(\bar{P}) = 2 \cdot \frac{5}{16} \log \frac{16}{5} + 2 \cdot \frac{3}{16} \log \frac{16}{3} = 1.049 + 0.906 = 1.954$ bits. Old code: $\frac{1}{2}(1.75 + 2.625) = 2.1875$. Fixed-length: 2. Known mission: 1.75. (c) The average waste is $H(\bar{P}, \bar{P}) - \frac{1}{2}H(P_1) - \frac{1}{2}H(P_2) = H(\bar{P}) - \frac{1}{2}(H(P_1) + H(P_2))$. With M uniform, X has pmf $\bar{P}$ and $H(X | M = m) = H(P_m)$, so this is $H(X) - H(X | M) = I(M; X) = 1.954 - 1.75 = 0.204$ bits: what one instruction reveals about the mission, and exactly the price of not knowing it. (d) Profiles (2, 2, 2, 2): cost 2. Profile (1, 2, 3, 3) with the short words on the two probable symbols: $\frac{5}{16} \cdot 1 + \frac{5}{16} \cdot 2 + \frac{3}{16} \cdot 3 + \frac{3}{16} \cdot 3 = \frac{33}{16} = 2.0625$. The fixed-length code wins. To approach 1.954, code blocks of instructions with lengths built for $\bar{P}$ (block coding theorem).

The log loss

Lecture Section 6

- B15. A three-class classifier. For a certain input the true label has pmf $p = (0.6, 0.3, 0.1)$. Five models report $q_1 = (0.6, 0.3, 0.1)$, $q_2 = (0.8, 0.15, 0.05)$, $q_3 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $q_4 = (0.98, 0.01, 0.01)$, $q_5 = (0.5, 0.4, 0.1)$. (a) Compute the expected log loss $H(p, q_i)$ of each. (b) Decompose each as $H(p) + D(p\|q_i)$. (c) q_4 puts the most probability on the most likely class, yet scores worst. Explain with one number. (d) A validation loss curve for such inputs has flattened at 1.30 bits per example. Is further training worthwhile? What is the perplexity at the floor?

Solution. (a) $H(p, q_1) = 1.295$, $H(p, q_2) = 0.6 \cdot 0.322 + 0.3 \cdot 2.737 + 0.1 \cdot 4.322 = 1.446$, $H(p, q_3) = \log 3 = 1.585$, $H(p, q_4) = 0.6 \cdot 0.029 + 0.3 \cdot 6.644 + 0.1 \cdot 6.644 = 2.675$, $H(p, q_5) = 1.329$ bits. (b) $H(p) = 1.295$; $D = 0, 0.151, 0.290, 1.380, 0.033$. The honest model q_1 has zero waste; q_5 , which merely shuffles 0.1 between the two likely classes, wastes little. (c) When class 2 or 3 occurs, which happens 40% of the time, q_4 charges $\log 100 = 6.64$ bits.

Confidence in the right answer is punished by the price of the wrong ones. (d) The floor is $H(p) = 1.295$; at most 0.005 bits per example remain to be gained, so no. Perplexity $2^{1.295} = 2.45$: the labels are as unpredictable as a choice among 2.45 equally likely classes.

Wordle

Lecture Section 7

- B16. Toy Wordle with three-letter words. $W = \{\text{ART, RAT, TAR, CAT, ACT, TIN, NIT, RIB}\}$, $N = 8$, answer A uniform on W . Colours: **G** right letter in the right place, **Y** right letter in the wrong place, **B** absent. (a) Write the pattern table $w \mapsto \pi_{\text{TIN}}(w)$ for the guess **TIN**, list the buckets and compute $H(\Pi)$. (b) The same for **RIB**. (c) Show that **CAT** gives eight distinct patterns, so $H(\Pi) = 3 = \log N$, the maximum possible. (d) **TAN** is not on the list. Compute its bucket sizes and entropy. Can any guess, on the list or off it, beat **CAT**? (e) Compute the expected number of guesses $\mathbb{E}[G]$ for the strategy “**CAT**, then the word it identifies” and for “**TIN**, then **CAT** if the bucket has four words, otherwise the identified word”. Do the entropy ranking and the score ranking agree?

Solution. The pattern tables of the four guesses, one row per possible answer:

answer w	$\pi_{\text{TIN}}(w)$	$\pi_{\text{RIB}}(w)$	$\pi_{\text{CAT}}(w)$	$\pi_{\text{TAN}}(w)$
ART	YBB	YBB	BYG	YYB
RAT	YBB	GBB	BGG	YGB
TAR	GBB	YBB	BGY	GGB
CAT	YBB	BBB	GGG	YGB
ACT	YBB	BBB	YYG	YYB
TIN	GGG	BGB	BBY	GBG
NIT	YGY	BGB	BBG	YBY
RIB	BGB	GGG	BBB	BBB
bucket sizes	4, 1, 1, 1, 1	2, 2, 2, 1, 1	1, ..., 1	2, 2, 1, 1, 1, 1
$H(\Pi)$	2	2.25	3	2.5

(a) **TIN**: bucket $\{\text{ART, RAT, CAT, ACT}\}$ and four singletons; $H(\Pi) = \frac{4}{8} \log 2 + 4 \cdot \frac{1}{8} \log 8 = 0.5 + 1.5 = 2$ bits. (b) **RIB**: buckets $\{\text{ART, TAR}\}$, $\{\text{CAT, ACT}\}$, $\{\text{TIN, NIT}\}$, $\{\text{RAT}\}$, $\{\text{RIB}\}$; $H = 3 \cdot \frac{2}{8} \cdot 2 + 2 \cdot \frac{1}{8} \cdot 3 = 2.25$ bits. (c) **CAT**: eight distinct patterns, $H = \log 8 = 3$. (d) **TAN**: sizes 2, 2, 1, 1, 1, 1; $H = 2 \cdot \frac{2}{8} \cdot 2 + 4 \cdot \frac{1}{8} \cdot 3 = 2.5$. No guess beats **CAT**: $H(\Pi) \leq \log N = 3$ always. (e) **CAT** first: $G = 1$ with probability $\frac{1}{8}$, else $G = 2$: $\mathbb{E}[G] = \frac{1}{8} + \frac{7}{8} \cdot 2 = 1.875$. **TIN** first: $\frac{1}{8}$ it is the answer ($G = 1$); $\frac{3}{8}$ a singleton ($G = 2$); $\frac{4}{8}$ the four-word bucket, where **CAT** produces four distinct patterns (**BYG**, **BGG**, **GGG**, **YYG**), so $G = 2$ with probability $\frac{1}{4}$ and 3 otherwise: $\mathbb{E}[G] = \frac{1}{8} + \frac{6}{8} + \frac{4}{8}(\frac{2}{4} + \frac{9}{4}) = 2.25$. Here more bits means fewer guesses; the lecture’s real-Wordle table shows the two rankings need not agree exactly.

- B17. Real Wordle, $N = 2315$ answers, 243 patterns. (a) A guess produces only five patterns, with bucket sizes 1000, 800, 400, 100, 15. Compute $H(\Pi)$ and the expected size of the remaining list, $\mathbb{E}[|\text{bucket of } \Pi|] = \sum_{\pi} |\text{bucket}| \cdot p_{\Pi}(\pi)$. (b) Another guess splits the list as evenly as 243 patterns allow: 128 buckets of size 10 and 115 of size 9. Compute $H(\Pi)$ and the expected remaining size. (c) Is the expected remaining size equal to $2^{\log N - H(\Pi)}$? Check with (a) and (b) and explain the discrepancy. (d) After a pattern whose bucket has 39 words, how many bits of uncertainty remain, and what is the largest amount of information the next guess could deliver?

Solution. (a) $H = \sum_s \frac{s}{N} \log \frac{N}{s} = 1.733$ bits, and the expected remaining size is

$$\sum_s s \cdot \frac{s}{N} = \frac{1000^2 + 800^2 + 400^2 + 100^2 + 15^2}{2315} = 782 \text{ words.}$$

(b) $H = 128 \cdot \frac{10}{2315} \log \frac{2315}{10} + 115 \cdot \frac{9}{2315} \log \frac{2315}{9} = 7.923$ bits, just under $\log 243 = 7.925$; expected remaining size $\frac{128 \cdot 100 + 115 \cdot 81}{2315} = 9.55$. (c) No. $2^{\log N - H}$ is 696 in (a) against 782, and 9.54 against 9.55 in (b) (nearly equal because the buckets are nearly equal). $\log N - H(\Pi) = \mathbb{E}[\log |\text{bucket}|]$, and the exponential of an average is at most the average of the exponentials (Jensen); the gap grows with the unevenness of the buckets. (d) $\log 39 = 5.29$ bits. A guess delivers $I(A; \Pi) = H(\Pi) \leq H(A) = \log 39$, so at most 5.29 bits; a guess achieving it would put each of the 39 words in its own bucket (possible in principle since $39 \leq 243$) and end the game on the following turn.

Joint and conditional entropy, mutual information

Lecture Section 8

- B18. Let X be the sum of two dice, $H(X) = 3.274$ bits. (a) The lecture exercises proved $H(g(X)) \leq H(X)$ for every function g . Show, from the chain rule for the pair $(X, g(X))$, that the gap is $H(X) - H(g(X)) = H(X | g(X))$, and say in words what this conditional entropy measures. (b) $g(X)$ = the parity of X . Compute $H(g(X))$ and $H(X | g(X))$. (c) $g(X) = X \bmod 3$. Compute $H(g(X))$ and $H(X | g(X))$ from (a). Then compute $H(X | g(X))$ directly, as the average of the three row entropies $H(X | g(X) = r)$, $r = 0, 1, 2$, and check that the two agree. (d) $g(X) = \mathbf{1}\{X \geq 7\}$. Compute $H(g(X))$. (e) Among all functions g of X with two values, which have the largest entropy, and what is it?

Solution. (a) $H(X, g(X)) = H(X) + H(g(X) | X) = H(X)$ since $g(X)$ is determined by X ; also $H(X, g(X)) = H(g(X)) + H(X | g(X))$. Subtracting, $H(X) - H(g(X)) = H(X | g(X))$: the bits of X that are lost by keeping only $g(X)$, i.e. the remaining uncertainty about X once $g(X)$ is known. (b) $\mathbb{P}(X \text{ even}) = \frac{1+3+5+5+3+1}{36} = \frac{1}{2}$: $H = 1$ bit; $H(X | \text{parity}) = 3.274 - 1 = 2.274$. (c) $X \bmod 3$: residue 0 is $X \in \{3, 6, 9, 12\}$ with weights $\frac{2+5+4+1}{36}$, residue 1 is $\{4, 7, 10\}$ with $\frac{3+6+3}{36}$, residue 2 is $\{2, 5, 8, 11\}$ with $\frac{1+4+5+2}{36}$: each $\frac{1}{3}$, so $H(g(X)) = \log 3 = 1.585$ and $H(X | X \bmod 3) = 3.274 - 1.585 = 1.689$. Directly: row 0 has conditional pmf $(\frac{2}{12}, \frac{5}{12}, \frac{4}{12}, \frac{1}{12})$, entropy 1.784; row 1 has $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, entropy 1.5; row 2 has $(\frac{1}{12}, \frac{4}{12}, \frac{5}{12}, \frac{2}{12})$, entropy 1.784. Average with weights $\frac{1}{3}$: $\frac{1}{3}(1.784 + 1.5 + 1.784) = 1.689$, agreeing. (d) $\mathbb{P}(X \geq 7) = \frac{21}{36}$, $H = h(\frac{7}{12}) = 0.980$ bits. (e) A two-valued g has $H = h(\mathbb{P}(g = 1)) \leq 1$, with equality iff $\mathbb{P}(g(X) = 1) = \frac{1}{2}$: any split of $\{2, \dots, 12\}$ into two sets of probability $\frac{1}{2}$, such as the parity in (b); 1 bit.

- B19. Two dice, $D = |Y - Z|$. (a) Write the pmf of D and compute $H(D)$. (b) Compute $H(D | Y = y)$ for each row y and hence $H(D | Y)$. (c) Compute $I(D; Y)$ and $H(Y | D)$. (d) Let $X = Y + Z$. Show that the pair (X, D) determines the unordered pair $\{Y, Z\}$ and nothing more, compute $H(X, D)$ from that, and hence $I(X; D)$. Explain where at least one of those bits comes from. (e) Compare $I(X; D)$ with $I(X; Y) = 0.689$ from the lecture.

Solution. (a) $D = 0, 1, 2, 3, 4, 5$ on 6, 10, 8, 6, 4, 2 cells: pmf $(6, 10, 8, 6, 4, 2)/36$; $H(D) = 2.441$ bits. (b) Row 1: $D = 0, 1, 2, 3, 4, 5$ once each, $H = \log 6 = 2.585$; row 6 the same. Row 2: $D = 1$ twice, 0, 2, 3, 4 once each: pmf $(\frac{2}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$, $H = 2.252$; row 5 the same. Rows 3, 4: $D = 1, 2$ twice each, 0, 3 once: $H = 1.918$. $H(D | Y) = \frac{1}{6}(2 \cdot 2.585 + 2 \cdot 2.252 + 2 \cdot 1.918) = 2.252$. (c) $I(D; Y) = 2.441 - 2.252 = 0.189$ bits; $H(Y | D) = H(Y) + H(D | Y) - H(D) = 2.585 + 2.252 - 2.441 = 2.396$. (d) $\{Y, Z\} = \{\frac{X+D}{2}, \frac{X-D}{2}\}$, and conversely X, D are functions of the unordered pair. The unordered pair takes 6 values (doubles) with probability $\frac{1}{36}$ and 15 values with probability $\frac{2}{36}$: $H(X, D) = \frac{6}{36} \log 36 + \frac{30}{36} \log 18 = 0.862 + 3.475 = 4.337$. Then $I(X; D) = H(X) + H(D) - H(X, D) = 3.274 + 2.441 - 4.337 = 1.379$ bits. One full

bit is the parity: X and D are always both even or both odd. (e) The difference tells you twice as much about the sum as the first die does (1.379 against 0.689): D constrains both the parity and the range of X ($D = 5$ forces $X = 7$), while Y leaves the sum uniform over six values.

- B20. A bit X with $\mathbb{P}(X = 1) = 0.3$ is sent down a wire that flips it with probability $\varepsilon = 0.2$, independently of X ; Y is received. (a) Write the 2×2 table of $p_{X,Y}$, rows $\{X = 0\}, \{X = 1\}$. (b) Compute $H(Y | X = 0)$, $H(Y | X = 1)$ and $H(Y | X)$. (c) Compute $\mathbb{P}(Y = 1)$, $H(Y)$ and $I(X; Y)$. (d) Compute $I(X; Y)$ for a uniform input and show that, for fixed $\varepsilon < \frac{1}{2}$, $I(X; Y)$ is maximised by the uniform input. (e) Compute $\mathbb{P}(X = 1 | Y = 1)$, $\mathbb{P}(X = 1 | Y = 0)$ and $H(X | Y)$, and check $I(X; Y) = H(X) - H(X | Y)$.

Solution. (a) Rows $\{X = 0\}$ (probability 0.7) and $\{X = 1\}$ (probability 0.3), each split 0.8/0.2 between the transmitted and the flipped value:

$p_{X,Y}$	$Y = 0$	$Y = 1$
$X = 0$	0.56	0.14
$X = 1$	0.06	0.24

(b) Each row, rescaled, is (0.8, 0.2) or (0.2, 0.8): $H(Y | X = x) = h(0.2) = 0.722$ for both, so $H(Y | X) = 0.722$. (c) $\mathbb{P}(Y = 1) = 0.14 + 0.24 = 0.38$; $H(Y) = h(0.38) = 0.958$; $I(X; Y) = 0.958 - 0.722 = 0.236$ bits. (d) Uniform input: $H(Y) = 1$, $I = 1 - h(0.2) = 0.278$. In general $\mathbb{P}(Y = 1) = a(1 - \varepsilon) + (1 - a)\varepsilon = \varepsilon + a(1 - 2\varepsilon)$ with $a = \mathbb{P}(X = 1)$, and $I = h(\mathbb{P}(Y = 1)) - h(\varepsilon)$; h is maximal at $\frac{1}{2}$, reached exactly when $a = \frac{1}{2}$. (e) $\mathbb{P}(X = 1 | Y = 1) = \frac{0.24}{0.38} = 0.632$, $\mathbb{P}(X = 1 | Y = 0) = \frac{0.06}{0.62} = 0.097$; $H(X | Y) = 0.38 h(0.632) + 0.62 h(0.097) = 0.645$; $H(X) = h(0.3) = 0.881$, and $0.881 - 0.645 = 0.236$.

Markov models, entropy rate, perplexity

Lecture Section 9

- B21. A binary text source is modelled as a Markov chain on $S = \{0, 1\}$ with transition matrix $Q = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$ (the chain of C12). (a) Find the stationary pmf π . (b) Compute the entropy rate $H(X_{i+1} | X_i)$ and the monographic entropy $H(\pi)$, and identify their difference. (c) Compute $H(X_1, X_2, X_3)$ for the chain started from π , once by the chain rule and once by listing the eight probabilities. (d) Give the perplexity of the Markov model, of the monographic model and of the uniform model. (e) What does a compressor built on the monographic model pay per symbol on this text, and how much of it is waste?

Solution. (a) Balance $0 \rightarrow 1$ against $1 \rightarrow 0$: $0.1\pi_0 = 0.3\pi_1$, so $\pi = (\frac{3}{4}, \frac{1}{4})$. (b) Rate $= \frac{3}{4}h(0.1) + \frac{1}{4}h(0.3) = 0.75 \cdot 0.469 + 0.25 \cdot 0.881 = 0.572$ bits per symbol; $H(\pi) = h(\frac{1}{4}) = 0.811$. The difference $0.239 = I(X_i; X_{i+1})$ is what the previous symbol tells you about the next. (c) Chain rule with the Markov property: $H(\pi) + 2 \cdot 0.572 = 1.955$. Listing: $\pi(a)Q(a, b)Q(b, c)$ over the eight triples is 0.6075, 0.0675, 0.0225, 0.0525, 0.0675, 0.0075, 0.0525, 0.1225 (for 000, 001, 010, 011, 100, 101, 110, 111), whose entropy is 1.955. (d) $2^{0.572} = 1.49$; $2^{0.811} = 1.75$; $2^1 = 2$. (e) Its cost is $\sum_x \pi(x) \log \frac{1}{\pi(x)} = H(\pi) = 0.811$ bits per symbol, since the symbol frequencies really are π ; the waste is $0.811 - 0.572 = 0.239$ bits per symbol, the mutual information of (b).

- B22. Perplexity, compression and a cipher. (a) A letter-level model scores 3.5 bits per letter on 27-symbol English. Give its perplexity and compare with Shannon's digraphic model (3.32 bits, perplexity 10). (b) A word-level model reports perplexity 40 per word. English words average 5.5 letters including the following space. Convert to bits per letter and

perplexity per letter. Which of the two models is better? (c) How many kilobytes does each model need to compress a text of 10^6 letters, and how many would the uniform model ($\log 27 = 4.75$ bits) need? (d) A Vigenère cipher with key length 10 has 26^{10} keys. Compute $H(K)$ and the unicity distance for $H(\text{English}) = 1.5$ and for 1 bit per letter (redundancy $R = \log 26 - H$). Why does a longer key need a longer ciphertext to be determined?

Solution. (a) $2^{3.5} = 11.3$: as unpredictable as a choice among 11.3 equally likely letters, slightly worse than the digraphic model's 10. (b) $\log 40 = 5.32$ bits per word, $5.32/5.5 = 0.968$ bits per letter, perplexity $2^{0.968} = 1.96$. The word model is much better: under one bit per letter, close to Shannon's long-context estimate. (c) $10^6 \cdot 0.968/8 = 121$ kB; $10^6 \cdot 3.5/8 = 437$ kB; $10^6 \cdot 4.75/8 = 594$ kB. (d) $H(K) = 10 \log 26 = 47.0$ bits. $R = 4.70 - 1.5 = 3.20$: $U = 47.0/3.20 = 14.7$ letters; $R = 3.70$: $U = 12.7$ letters. Each ciphertext letter leaks R bits about the key, so a key with more bits of entropy needs proportionally more letters before none is left.

Part C: Stochastic Processes

Notation as in the Stochastic Processes lecture: $\xi_1, \xi_2, \dots$ are the ± 1 steps, $S_n = \xi_1 + \dots + \xi_n$, and $\mathbb{P}_i, \mathbb{E}_i$ refer to a start at i . $\mathbf{P} = (p(i, j))$ is the transition matrix, row i the conditional pmf of the next state given the current state i ; $p^{(n)}(i, j)$ are the entries of $\mathbf{P}^n$; π is a stationary distribution; T_A is the hitting time of A , $h_i^A = \mathbb{P}_i(T_A < \infty)$, $k_i^A = \mathbb{E}_i[T_A]$; $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$ is the fundamental matrix. Φ is the $N(0, 1)$ cdf; values used below:

x	0.100	0.500	0.577	0.674	0.707	0.905	1.000	1.155	1.225	
$\Phi(x)$	0.540	0.691	0.718	0.750	0.760	0.817	0.841	0.876	0.890	$\Phi(1.414) = 0.921$.

Processes and the random walk

Lecture batteries 1–2

- C1. Three fair coin tosses, $\xi_t = \pm 1$, $S_t = \xi_1 + \dots + \xi_t$: the eight-row table of the lecture. (a) Write the joint pmf of (S_1, S_3) as a table and check that S_1 and S_3 are not independent. (b) Read $\mathbb{E}[S_3 \mid S_1 = s]$ and $\mathbb{E}[S_3 \mid S_2 = s]$ off the rows, for every s , and explain the answers by independent increments. (c) Compute $\mathbb{E}[S_3^2 \mid S_2 = s]$ for $s = -2, 0, 2$ and write $\mathbb{E}[S_3^2 \mid S_2]$ as a random variable.

Solution. (a) Each row has probability $\frac{1}{8}$; (S_1, S_3) takes $(1, 3)$ on HHH, $(1, 1)$ on HHT, HTH, $(1, -1)$ on HTT, $(-1, 1)$ on THH, $(-1, -1)$ on THT, TTH, $(-1, -3)$ on TTT:

p_{S_1, S_3}	$S_3 = -3$	$S_3 = -1$	$S_3 = 1$	$S_3 = 3$
$S_1 = 1$	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$
$S_1 = -1$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0

So $\mathbb{P}(S_1 = 1, S_3 = 1) = \frac{2}{8} \neq \mathbb{P}(S_1 = 1)\mathbb{P}(S_3 = 1) = \frac{1}{2} \cdot \frac{3}{8}$. (b) Rows with $S_1 = 1$: $S_3 = 3, 1, 1, -1$, average 1; with $S_1 = -1$: average -1 . Rows with $S_2 = 2$: $S_3 = 3, 1$, average 2; $S_2 = 0$: $S_3 = 1, -1, 1, -1$, average 0; $S_2 = -2$: average -2 . In each case $\mathbb{E}[S_3 \mid S_m = s] = s$, because $S_3 = S_m + (\xi_{m+1} + \dots + \xi_3)$ and the increment is independent of S_m with mean 0. (c) S_3^2 on the rows with $S_2 = 2$: 9, 1, average 5; $S_2 = 0$: 1, 1, 1, 1, average 1; $S_2 = -2$: 5. So $\mathbb{E}[S_3^2 \mid S_2 = s] = s^2 + 1$ and $\mathbb{E}[S_3^2 \mid S_2] = S_2^2 + 1$: the square of the known position plus the variance of one step.

- C2. A walk with $p = 0.55$ (a slightly favourable game), $n = 100$ steps. (a) Compute $\mathbb{E}[S_{100}]$, $\text{Var}(S_{100})$ and the standard deviation. (b) Compute $\mathbb{P}(S_{100} = 10)$ and $\mathbb{P}(S_{100} = 11)$. (c) Approximate $\mathbb{P}(S_{100} > 0)$ by the normal distribution (use $\{S_{100} \geq 2\}$ and the midpoint 1

as the cut). (d) Compute $\text{Cov}(S_{30}, S_{100})$ and the correlation of S_{30} and S_{100} . (e) Fair walk: compute $\mathbb{P}(S_8 = 0)$, $\mathbb{P}(S_8 = 4)$ and $\mathbb{P}(S_9 = 0)$.

Solution. (a) $\mathbb{E}[S_{100}] = 100(0.55 - 0.45) = 10$; $\text{Var}(S_{100}) = 4 \cdot 100 \cdot 0.55 \cdot 0.45 = 99$; standard deviation 9.95. (b) $S_{100} = 10$ iff $H_{100} = 55$: $\binom{100}{55} 0.55^{55} 0.45^{45} = 0.080$. $S_{100} = 11$ is impossible: S_{100} is even. (c) $\mathbb{P}(S_{100} \geq 2) \approx \mathbb{P}(N(10, 99) > 1) = 1 - \Phi(\frac{1-10}{9.95}) = \Phi(0.905) = 0.817$; the exact binomial sum is 0.817. (d) $S_{100} = S_{30} + (S_{100} - S_{30})$ with the increment independent of S_{30} , so $\text{Cov}(S_{30}, S_{100}) = \text{Var}(S_{30}) = 4 \cdot 30 \cdot 0.55 \cdot 0.45 = 29.7$; correlation $29.7/\sqrt{29.7 \cdot 99} = \sqrt{30/100} = 0.548$. (e) $\mathbb{P}(S_8 = 0) = \binom{8}{4}/256 = \frac{70}{256} = \frac{35}{128}$; $\mathbb{P}(S_8 = 4) = \binom{8}{6}/256 = \frac{28}{256} = \frac{7}{64}$; $\mathbb{P}(S_9 = 0) = 0$ by parity.

- C3. Fair walk, $M_8 = \max_{k \leq 8} S_k$. (a) Compute $\mathbb{P}(M_8 \geq 2)$ by the reflection principle. (b) Compute $\mathbb{P}(M_8 = 0)$. (c) Tabulate $\mathbb{P}(M_8 \geq m)$ for $m = 1, \dots, 8$ and deduce the pmf of M_8 . (d) Compute $\mathbb{E}[M_8] = \sum_{m \geq 1} \mathbb{P}(M_8 \geq m)$. What would the head-on computation require?

Solution. (a) $\mathbb{P}(M_8 \geq 2) = 2\mathbb{P}(S_8 > 2) + \mathbb{P}(S_8 = 2) = 2\mathbb{P}(S_8 \geq 4) + \mathbb{P}(S_8 = 2) = 2 \cdot \frac{28+8+1}{256} + \frac{56}{256} = \frac{130}{256} = \frac{65}{128}$. (b) $\mathbb{P}(M_8 = 0) = \mathbb{P}(S_8 = 0) + \mathbb{P}(S_8 = 1) = \frac{70}{256} + 0 = \frac{35}{128}$: over a quarter of all paths never rise above their start. (c) $\mathbb{P}(M_8 \geq m)$ for $m = 1, \dots, 8$: $\frac{93}{128}, \frac{65}{128}, \frac{37}{128}, \frac{23}{128}, \frac{9}{128}, \frac{5}{128}, \frac{1}{128}, \frac{1}{256}$. Differences give the pmf of M_8 on $0, 1, \dots, 8$: $\frac{70}{256}, \frac{56}{256}, \frac{28}{256}, \frac{8}{256}, \frac{1}{256}, \frac{1}{256}, \frac{1}{256}, \frac{1}{256}$, which is $\mathbb{P}(S_8 = m) + \mathbb{P}(S_8 = m+1)$ as in the lecture's corollary. (d) $\mathbb{E}[M_8] = \frac{186+130+74+46+18+10+2+1}{256} = \frac{467}{256} = 1.824$. Head-on: track the running maximum along each of the 256 paths.

- C4. A fair walk on $\{0, 1, \dots, a\}$ with a reflecting wall: from 0 the next position is 1 with probability one; from $0 < i < a$ it moves to $i \pm 1$ with probability $\frac{1}{2}$ each; a is absorbing. Let $m_i = \mathbb{E}_i[\text{time to reach } a]$. (a) Write the first-step equations for $m_0, \dots, m_a$. (b) Solve them for $a = 3$. (c) Show that $m_i = a^2 - i^2$ for every a , and explain the agreement with the lecture's exit time from $(-a, a)$.

Solution. (a) $m_a = 0$; $m_0 = 1 + m_1$; $m_i = 1 + \frac{1}{2}m_{i-1} + \frac{1}{2}m_{i+1}$ for $0 < i < a$. (b) $a = 3$: $m_3 = 0$, $m_2 = 1 + \frac{1}{2}m_1$, $m_1 = 1 + \frac{1}{2}m_0 + \frac{1}{2}m_2$, $m_0 = 1 + m_1$. Substituting, $m_1 = 1 + \frac{1}{2}(1 + m_1) + \frac{1}{2}(1 + \frac{1}{2}m_1)$, so $\frac{1}{4}m_1 = 2$: $m_1 = 8$, $m_0 = 9$, $m_2 = 5$. (c) $m_i = a^2 - i^2$: at $i = a$ it is 0; at $0 < i < a$, $1 + \frac{1}{2}(a^2 - (i-1)^2) + \frac{1}{2}(a^2 - (i+1)^2) = a^2 - i^2$; at 0, $1 + (a^2 - 1) = a^2$. Uniqueness as in the lecture. If X_n is the unrestricted fair walk started at i , then $|X_n|$ is exactly this reflected walk, and $|X_n|$ reaches a when X_n leaves $(-a, a)$: the same random time, so the same expectation $a^2 - i^2$.

Brownian motion

Lecture battery 3

- C5. Let B be a standard Brownian motion. (a) Give the distribution of $B_3 - B_1$ and compute $\mathbb{P}(B_3 - B_1 > 2)$. (b) Compute $\mathbb{P}(B_4 > 1 \mid B_1 = 2)$ and $\mathbb{P}(B_4 > 1 \mid B_1 = -1)$ by splitting into increments. (c) Write $\mathbb{E}[B_4 \mid B_1]$ and $\mathbb{E}[B_4^2 \mid B_1]$ as random variables. (d) Now condition the other way. Explain by symmetry between the three increments $B_1, B_2 - B_1, B_3 - B_2$ why $\mathbb{E}[B_1 \mid B_3] = B_3/3$. Then, by completing the square in the joint pdf of (B_1, B_3) , show that given $B_3 = x$, $B_1 \sim N(\frac{x}{3}, \frac{2}{3})$, and compute $\mathbb{P}(B_1 > 0 \mid B_3 = 3)$.

Solution. (a) $B_3 - B_1 \sim N(0, 2)$; $\mathbb{P}(B_3 - B_1 > 2) = 1 - \Phi(2/\sqrt{2}) = 1 - \Phi(1.414) = 0.079$. (b) $B_4 = B_1 + (B_4 - B_1)$ with $B_4 - B_1 \sim N(0, 3)$ independent of B_1 . Given $B_1 = 2$: $B_4 \sim N(2, 3)$, $\mathbb{P}(B_4 > 1) = \mathbb{P}(N(0, 1) > -1/\sqrt{3}) = \Phi(0.577) = 0.718$. Given $B_1 = -1$: $B_4 \sim N(-1, 3)$, $\mathbb{P} = 1 - \Phi(2/\sqrt{3}) = 1 - \Phi(1.155) = 0.124$. (c) $\mathbb{E}[B_4 \mid B_1] = B_1 + \mathbb{E}[B_4 - B_1] = B_1$; $\mathbb{E}[B_4^2 \mid B_1] = B_1^2 + 2B_1\mathbb{E}[B_4 - B_1] + \mathbb{E}[(B_4 - B_1)^2] = B_1^2 + 3$. (d) The three increments are i.i.d. $N(0, 1)$ and B_3 is their sum, so given B_3 each has the same conditional mean and the three means add to B_3 : $\mathbb{E}[B_1 \mid B_3] = B_3/3$ (as $\mathbb{E}[S_5 \mid S_8] = \frac{5}{8}S_8$ in the Conditional Expectation

lecture, battery 4, exercise 4). Joint pdf: $f(x, y) = \frac{1}{2\pi\sqrt{2}}e^{-x^2/2}e^{-(y-x)^2/4}$. As a function of x the exponent is $-\frac{x^2}{2} - \frac{(y-x)^2}{4} = -\frac{3}{4}(x - \frac{y}{3})^2 + \text{const}(y)$, so $f_{B_1|B_3}(x | y) \propto e^{-(x-y/3)^2/(2 \cdot 2/3)}$; $N(\frac{y}{3}, \frac{2}{3})$. Then $\mathbb{P}(B_1 > 0 | B_3 = 3) = \mathbb{P}(N(1, \frac{2}{3}) > 0) = \Phi(1/\sqrt{2/3}) = \Phi(1.225) = 0.890$.

- C6. Use $\text{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t] = \min(s, t)$ throughout. (a) Compute $\text{Var}(B_1 + B_2 + B_3)$. (b) Compute $\text{Var}(B_2 - 2B_1)$ and $\text{Cov}(B_2 - 2B_1, B_1)$. For which constant c is $B_2 - cB_1$ uncorrelated with B_1 ? Which property of Brownian motion says that for this c it is in fact independent of B_1 ? (c) Show $\text{Corr}(B_s, B_t) = \sqrt{s/t}$ for $s < t$, and evaluate it for $(1, 4)$ and $(1, 100)$. (d) Let $X_t = B_{t+1} - B_t$ for $t \geq 0$. Show that $X_t \sim N(0, 1)$ for every t and that $\text{Cov}(X_s, X_t) = \max(0, 1 - |t - s|)$. In what sense is X stationary although B is not?

Solution. (a) $\sum_{s,t \in \{1,2,3\}} \min(s, t) = (1+2+3) + 2(1+1+2) = 14$. (b) $\text{Var}(B_2 - 2B_1) = \text{Var}(B_2) - 4\text{Cov}(B_2, B_1) + 4\text{Var}(B_1) = 2 - 4 + 4 = 2$; $\text{Cov}(B_2 - 2B_1, B_1) = 1 - 2 = -1 \neq 0$, so not independent. $\text{Cov}(B_2 - cB_1, B_1) = 1 - c = 0$ iff $c = 1$: the only such combination is the increment $B_2 - B_1$, and property (ii) (independent increments) says that this increment is independent of $B_1 = B_1 - B_0$, not merely uncorrelated. (c) $\frac{\min(s,t)}{\sqrt{st}} = \frac{s}{\sqrt{st}} = \sqrt{s/t}$; $\sqrt{1/4} = 0.5$ and $\sqrt{1/100} = 0.1$. (d) X_t is an increment over an interval of length 1, so $N(0, 1)$. For $s \leq t$: $\text{Cov}(X_s, X_t) = \min(s+1, t+1) - \min(s+1, t) - \min(s, t+1) + \min(s, t) = (s+1) - \min(s+1, t) - s + s = s+1 - \min(s+1, t)$, which is $1 - (t - s)$ if $t \leq s+1$ and 0 if $t > s+1$. Every X_t has the same distribution and the covariance depends only on $t - s$: the process of unit increments looks the same at every time, while $\text{Var}(B_t) = t$ grows.

- C7. The reflection principle for Brownian motion: $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a)$. Let T_1 be the first time B reaches 1. (a) Compute $\mathbb{P}(\max_{s \leq 4} B_s \geq 2)$. (b) Write $\mathbb{P}(T_1 \leq t)$ in terms of Φ and evaluate it at $t = 1, 4, 100$. (c) Find the median of T_1 (the t with $\mathbb{P}(T_1 \leq t) = \frac{1}{2}$). (d) By scaling, what is the median of T_a ? (e) The lecture showed $\mathbb{E}[T_1] = \infty$. Reconcile this with a median of about 2.

Solution. (a) $2\mathbb{P}(B_4 \geq 2) = 2\mathbb{P}(N(0, 1) \geq 1) = 2(1 - \Phi(1)) = 0.317$. (b) $\{T_1 \leq t\} = \{\max_{s \leq t} B_s \geq 1\}$, so $\mathbb{P}(T_1 \leq t) = 2(1 - \Phi(1/\sqrt{t}))$: 0.317 at $t = 1$; $2(1 - \Phi(0.5)) = 0.617$ at $t = 4$; $2(1 - \Phi(0.1)) = 0.920$ at $t = 100$. (c) $2(1 - \Phi(1/\sqrt{t})) = \frac{1}{2}$ iff $\Phi(1/\sqrt{t}) = \frac{3}{4}$ iff $1/\sqrt{t} = 0.674$: median $t = 1/0.674^2 = 2.20$. (d) $\frac{1}{a}B_{a^2t}$ is a Brownian motion, and it reaches 1 exactly when B reaches a , at time a^2 times smaller; so T_a has the distribution of a^2T_1 and median $2.20a^2$. (e) Half the paths reach 1 within 2.2 time units, but the tail is heavy: $\mathbb{P}(T_1 > t) = 2\Phi(1/\sqrt{t}) - 1 \approx \sqrt{2/(\pi t)}$, and $\int^\infty t^{-1/2} dt = \infty$. A typical wait is short; the rare long waits are long enough to make the mean infinite.

Risk of ruin

Lecture batteries 4–5

- C8. A gambler with £20 plays £1 rounds until she has £50 or nothing. (a) Fair game: compute the probability of reaching £50 and the expected number of rounds. (b) $p = 0.48$: the same, to three decimals. (c) She switches to £10 rounds. Redo (a) and (b) (a walk from 2 to 5); verify the unfair probability by solving the four first-step equations directly rather than by the formula. (d) State in one sentence what the stake changes and what it does not.

Solution. (a) $h_{20} = \frac{20}{50} = 0.4$; $D_{20} = 20 \cdot 30 = 600$ rounds. (b) $r = \frac{0.52}{0.48} = \frac{13}{12}$; $h_{20} = \frac{r^{20}-1}{r^{50}-1} = 0.074$; $D_{20} = \frac{20}{0.04} - \frac{50}{0.04} \cdot \frac{1-r^{20}}{1-r^{50}} = 500 - 1250 \cdot 0.0737 = 408$ rounds. (c) Fair: $h_2 = \frac{2}{5} = 0.4$, $D_2 = 2 \cdot 3 = 6$ rounds. Unfair: $h_0 = 0$, $h_5 = 1$, $h_i = 0.48h_{i+1} + 0.52h_{i-1}$ for $i = 1, \dots, 4$; eliminating, $h_2 = \frac{43200}{122461} = 0.353$ (the formula gives $\frac{r^2-1}{r^5-1}$, the same), and $D_2 = 5.90$ rounds. (d) In the fair game the stake changes only the duration; in the unfair game bigger stakes mean fewer rounds for the edge to compound, and the chance of reaching the target rises from 0.074 to 0.353.

- C9. Red at roulette: European wheels have one zero ($p = \frac{18}{37}$), American wheels two ($p = \frac{18}{38}$). A player starts with £100 and bets £1 a spin. (a) Compute the probability of reaching £200 on each wheel, and the ratio. (b) Compute the expected number of spins on each wheel. (c) European wheel, modest target £110: compute h_{100} and $\mathbb{E}[T]$, and check the identity mean gain $= Nh_i - i = (p - q) \mathbb{E}_i[T]$ numerically. (d) With no target, ruin is certain on both wheels. Compute the expected number of spins until ruin on each.

Solution. (a) *European:* $r = \frac{19}{18}$, $h_{100} = \frac{r^{100}-1}{r^{200}-1} = 0.00447$. *American:* $r = \frac{10}{9}$, $h_{100} = 0.0000266$. A second zero divides the chance by about 170. (b) $D_{100} = \frac{100}{q-p} - \frac{200}{q-p} \cdot \frac{1-r^{100}}{1-r^{200}}$: *European* $q - p = \frac{1}{37}$, $D = 3667$ spins; *American* $q - p = \frac{2}{38}$, $D = 1900$ spins. (c) $h_{100} = \frac{r^{100}-1}{r^{110}-1} = 0.581$, $\mathbb{E}[T] = 1334$ spins. Mean gain $110 \cdot 0.581 - 100 = -36.1$; and $(p - q)\mathbb{E}[T] = -\frac{1}{37} \cdot 1334 = -36.1$. (d) $\frac{i}{q-p}$: *European* $100 \cdot 37 = 3700$ spins; *American* $100 \cdot 19 = 1900$ spins.

- C10. Bold play. $p = 0.4$, fortune £3, target £10. (a) Timid play (£1 per round): compute h_3 and $\mathbb{E}_3[T]$. (b) Bold play: each round stake $\min(i, 10 - i)$, so the fortune moves from 3 to 6 or 0, from 6 to 10 or 2, from 2 to 4 or 0, from 4 to 8 or 0, from 8 to 10 or 6. Draw the chain on $\{0, 2, 3, 4, 6, 8, 10\}$ and write the hitting system for $h_i = \mathbb{P}_i(\text{reach } 10)$. (c) Solve it and compare h_3 with (a). (d) Compute the expected number of bets under bold play from £3. (e) Repeat (c) for the fair game $p = \frac{1}{2}$. Why is the answer the same as for timid play there?

Solution. (a) $r = 1.5$: $h_3 = \frac{1.5^3-1}{1.5^{10}-1} = 0.042$; $\mathbb{E}_3[T] = \frac{3}{0.2} - \frac{10}{0.2} \cdot \frac{1-1.5^3}{1-1.5^{10}} = 12.9$ rounds. (b) $h_0 = 0$, $h_{10} = 1$, $h_3 = ph_6$, $h_6 = p + qh_2$, $h_2 = ph_4$, $h_4 = ph_8$, $h_8 = p + qh_6$. (c) From $h_8 = p + qh_6$ and $h_6 = p + qh_2 = p + qp^2h_8$: $h_8 = p + qp + q^2p^2h_8$, so $h_8 = \frac{p(1+q)}{1-q^2p^2} = \frac{0.64}{0.9424} = \frac{400}{589} = 0.679$; then $h_6 = \frac{274}{589} = 0.465$, $h_4 = \frac{160}{589} = 0.272$, $h_2 = \frac{64}{589} = 0.109$, and $h_3 = ph_6 = \frac{548}{2945} = 0.186$: four and a half times the timid 0.042. (d) $k_0 = k_{10} = 0$, $k_3 = 1 + pk_6$, $k_6 = 1 + qk_2$, $k_2 = 1 + pk_4$, $k_4 = 1 + pk_8$, $k_8 = 1 + qk_6$: $k_3 = \frac{1073}{589} = 1.82$ bets, against 12.9 rounds of timid play. (e) With $p = \frac{1}{2}$ the system gives $h_3 = \frac{3}{10}$, the timid value. A fair game preserves expected fortune whatever the stakes, so $\mathbb{E}[X_T] = 3 = 10h_3$ for every strategy that stops at 0 or 10.

Markov chains, paths and n -step transitions

Lecture batteries 6–7

- C11. A taxi works three zones A, B, C. The zone where a trip ends, hence where the next trip starts, depends only on where the current trip started:

$$\mathbf{P} = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix} \quad (\text{rows and columns in the order A, B, C}).$$

X_n is the starting zone of trip n . (a) Check that $\mathbf{P}$ is stochastic and say in words what row B means. (b) Compute $\mathbb{P}(X_1 = C, X_2 = B \mid X_0 = A)$. (c) Compute row A of $\mathbf{P}^2$ by hand and read off $\mathbb{P}(X_2 = C \mid X_0 = A)$. Which three two-trip sequences does that entry add up? (d) X_0 is A or B with probability $\frac{1}{2}$ each. Compute the distributions of X_1 and X_2 . (e) Given $X_1 = B$, does the value of X_0 change the distribution of X_2 ? Which property says so?

Solution. (a) Entries non-negative, every row sums to 1. Row B: after a trip starting in B, the next trip starts in A with probability 0.2, in B with probability 0.6, in C with probability 0.2. (b) $p(A, C)p(C, B) = 0.2 \cdot 0.3 = 0.06$. (c) $(0.5, 0.3, 0.2)\mathbf{P}^2 = (0.25 + 0.06 + 0.06, 0.15 + 0.18 + 0.06, 0.10 + 0.06 + 0.08) = (0.37, 0.39, 0.24)$; $\mathbb{P}(X_2 = C \mid X_0 = A) =$

$0.24 = p(A, A)p(A, C) + p(A, B)p(B, C) + p(A, C)p(C, C)$: the sequences AAC, ABC, ACC. (d) $\lambda = (\frac{1}{2}, \frac{1}{2}, 0)$: $\lambda \mathbf{P} = (0.35, 0.45, 0.20)$ and $\lambda \mathbf{P}^2 = (0.35, 0.45, 0.20) \mathbf{P} = (0.325, 0.435, 0.24)$. (e) No: $\mathbb{P}(X_2 = j \mid X_1 = B, X_0 = i) = p(B, j)$ for every i . That is the Markov property.

- C12. The two-state chain $\mathbf{P} = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$ on $\{0, 1\}$ (the text source of B21). (a) Find the eigenvalues of $\mathbf{P}$ and the stationary distribution π . (b) Show, by induction or from the lecture's formula for two-state chains, that $\mathbf{P}^n = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix} + 0.6^n \begin{pmatrix} \frac{1}{4} & -\frac{1}{4} \\ -\frac{3}{4} & \frac{3}{4} \end{pmatrix}$, and check $n = 1, 2$. (c) Compute $p^{(3)}(0, 0)$ and $\mathbb{P}(X_3 = 0 \mid X_0 = 1)$ exactly. (d) Find the smallest n for which every entry of $\mathbf{P}^n$ is within 0.01 of π .

Solution. (a) Trace 1.6, so the eigenvalues are 1 and 0.6. $\pi \mathbf{P} = \pi$: $0.1\pi_0 = 0.3\pi_1$, $\pi = (\frac{3}{4}, \frac{1}{4})$. (b) $n = 1$: $\frac{3}{4} + 0.6 \cdot \frac{1}{4} = 0.9$, $\frac{1}{4} - 0.6 \cdot \frac{1}{4} = 0.1$, $\frac{3}{4} - 0.6 \cdot \frac{3}{4} = 0.3$, $\frac{1}{4} + 0.6 \cdot \frac{3}{4} = 0.7$. Induction: multiplying the first matrix by $\mathbf{P}$ leaves it unchanged (its rows are π), and multiplying the second by $\mathbf{P}$ multiplies it by the eigenvalue 0.6 (its rows are proportional to $(1, -1)$, and $(1, -1)\mathbf{P} = (0.6, -0.6)$). $n = 2$: $\mathbf{P}^2 = \begin{pmatrix} 0.84 & 0.16 \\ 0.48 & 0.52 \end{pmatrix}$. (c) $p^{(3)}(0, 0) = \frac{3}{4} + \frac{1}{4} \cdot 0.216 = 0.804 = \frac{201}{250}$; $p^{(3)}(1, 0) = \frac{3}{4} - \frac{3}{4} \cdot 0.216 = 0.588 = \frac{147}{250}$. (d) The largest deviation is $\frac{3}{4} \cdot 0.6^n$: 0.0126 at $n = 8$, 0.0076 at $n = 9$. So $n = 9$.

- C13. The vowel/consonant chain of both lectures: $S = \{V, C\}$, $\mathbf{P} = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}$, $\pi = (\frac{5}{13}, \frac{8}{13})$.

(a) Compute $\mathbf{P}^2$ and $\mathbf{P}^3$. (b) Compute the expected number of vowels among the next three letters, starting after a consonant and starting after a vowel, by linearity column by column; compare with the long-run value $3\pi_V$. (c) Compute $\mathbb{P}(X_1 = V, X_2 = V \mid X_0 = C)$ and compare with the stationary probability of the digraph VV used in the Entropy lecture. (d) Compute the expected number of letters until the first consonant, starting from V ; then the expected number of letters until the pattern VC is first completed, from V and from C . (e) Compute the mean return time to C from π and confirm it by a first-step argument.

Solution. (a) $\mathbf{P}^2 = \begin{pmatrix} 0.44 & 0.56 \\ 0.35 & 0.65 \end{pmatrix}$, $\mathbf{P}^3 = \begin{pmatrix} 0.368 & 0.632 \\ 0.395 & 0.605 \end{pmatrix}$. (b) $\mathbb{E}_C[\sum_{n=1}^3 \mathbf{1}\{X_n = V\}] = \sum_{n=1}^3 p^{(n)}(C, V) = 0.5 + 0.35 + 0.395 = 1.245$; from V : $0.2 + 0.44 + 0.368 = 1.008$; long run $3 \cdot \frac{5}{13} = 1.154$, in between. (c) $p(C, V)p(V, V) = 0.5 \cdot 0.2 = 0.1$; stationary $\pi_V p(V, V) = \frac{5}{13} \cdot 0.2 = \frac{1}{13} = 0.077$. A consonant just before makes a vowel, hence VV , likelier than average. (d) From V each letter is a consonant with probability 0.8: geometric, mean $1/0.8 = 1.25$. Pattern VC : from V the pattern completes at the first consonant, $k_V = 1.25$; from C , $k_C = 1 + 0.5k_V + 0.5k_C$, so $k_C = 2 + k_V = 3.25$. (e) $1/\pi_C = \frac{13}{8} = 1.625$. First step from C : with probability 0.5 back at once, with probability 0.5 at V and then 1.25 more letters: $1 + 0.5 \cdot 1.25 = 1.625$.

Stationary distributions

Lecture battery 8

- C14. Random walks on graphs: $p(i, j) = 1/d(i)$ for each neighbour j of i . (a) The path $0-1-2-3$. Find π by detailed balance and verify the balance equation on the edges $(0, 1)$ and $(1, 2)$. (b) The star with centre 0 and leaves 1, 2, 3. Find π ; compute the mean return time to a leaf from π and confirm it by a first-step argument. (c) The Ehrenfest urn with $N = 3$ balls. Find π by detailed balance. (d) The biased walk on the cycle $\mathbb{Z}_3$: $p(i, i+1) = \frac{2}{3}$, $p(i, i-1) = \frac{1}{3}$. Show that the uniform distribution is stationary but that detailed balance fails. What does the chain do that a reversible chain cannot?

Solution. (a) $\pi_i \propto d(i)$: $\pi = (1, 2, 2, 1)/6$. Edge $(0, 1)$: $\frac{1}{6} \cdot 1 = \frac{2}{6} \cdot \frac{1}{2}$. Edge $(1, 2)$: $\frac{2}{6} \cdot \frac{1}{2} = \frac{2}{6} \cdot \frac{1}{2}$. (b) Degrees $(3, 1, 1, 1)$: $\pi = (3, 1, 1, 1)/6$; mean return time to leaf 1 is $1/\pi_1 = 6$. First step: from the leaf to the centre (1 step); from the centre the walk picks leaf 1 with probability $\frac{1}{3}$, otherwise makes a wasted excursion of 2 steps and is back at the centre; the number of wasted excursions is geometric with mean 2; total $1 + 2 \cdot 2 + 1 = 6$. (c) $\pi_i \frac{3-i}{3} = \pi_{i+1} \frac{i+1}{3}$, so $\pi_1 = 3\pi_0$, $\pi_2 = \pi_1$, $\pi_3 = \frac{1}{3}\pi_2$: $\pi = (1, 3, 3, 1)/8 = \binom{3}{i} 2^{-3}$. (d) Each column of $\mathbf{P}$ sums to $\frac{2}{3} + \frac{1}{3} = 1$, so $\sum_i \frac{1}{3} p(i, j) = \frac{1}{3}$: uniform π is stationary. Detailed balance: $\pi_0 p(0, 1) = \frac{2}{9} \neq \frac{1}{9} = \pi_1 p(1, 0)$. The chain has a net circulation ($\frac{1}{9}$ per step anticlockwise around the cycle); run backwards it circulates the other way, so its law is not the same. A reversible chain has zero net flow on every edge.

- C15. The taxi chain of C11. (a) Solve $\pi \mathbf{P} = \pi$, $\sum_i \pi_i = 1$ exactly. (b) $\mathbf{P}^{10}$ has every row equal to $(0.3214, 0.4286, 0.2500)$ to four places. Confirm that this agrees with (a). (c) Test detailed balance on the pair (A, B) . Is the chain reversible? (d) The chain is irreducible and aperiodic. What do the convergence theorem and the ergodic theorem each say about $\pi_C = \frac{1}{4}$?

Solution. (a) Columns A and C: $0.5\pi_A + 0.2\pi_B + 0.3\pi_C = \pi_A$ and $0.2\pi_A + 0.2\pi_B + 0.4\pi_C = \pi_C$, i.e. $5\pi_A = 2\pi_B + 3\pi_C$ and $3\pi_C = \pi_A + \pi_B$. With $\pi_A + \pi_B + \pi_C = 1$ the second gives $\pi_C = \frac{1}{4}$, then $5\pi_A = 2\pi_B + \frac{3}{4}$ and $\pi_A + \pi_B = \frac{3}{4}$ give $\pi = (\frac{9}{28}, \frac{3}{7}, \frac{1}{4}) = (0.3214, 0.4286, 0.25)$. (b) $\frac{9}{28} = 0.32143$, $\frac{3}{7} = 0.42857$: the rows of $\mathbf{P}^{10}$ are π to four places, as the convergence theorem predicts. (c) $\pi_A p(A, B) = \frac{9}{28} \cdot 0.3 = \frac{27}{280}$ but $\pi_B p(B, A) = \frac{3}{7} \cdot 0.2 = \frac{24}{280}$. Not reversible (unlike the weather chain of the lecture): more trips go $A \rightarrow B$ than $B \rightarrow A$ in the long run. (d) Convergence: $\mathbb{P}(X_n = C) \rightarrow \frac{1}{4}$ whatever the starting zone. Ergodic: along a single sequence of trips, the fraction starting in C tends to $\frac{1}{4}$ with probability one, and consecutive C-starts are $1/\pi_C = 4$ trips apart on average.

- C16. PageRank with damping $d = 0.85$. (a) Three pages: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow A$. Write $\mathbf{H}$ and $\mathbf{P} = 0.85\mathbf{H} + 0.05\mathbf{J}$. (b) Solve $\pi \mathbf{P} = \pi$ and rank the pages. A and B each have exactly one in-link; why does A rank higher? (c) Find π without damping ($d = 1$). (d) Add a page D with links $C \rightarrow D$ (so C now links to A and D) and $D \rightarrow D$ only. Find π without damping, and with $d = 0.85$. What does D do to the surfer, and how much does damping help?

Solution. (a) $\mathbf{H} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} 0.05 & 0.475 & 0.475 \\ 0.05 & 0.05 & 0.9 \\ 0.9 & 0.05 & 0.05 \end{pmatrix}$. (b) $\pi \mathbf{P} = \pi$ with $\sum \pi = 1$:

$\pi_A = 0.05 + 0.85\pi_C$, $\pi_B = 0.05 + 0.425\pi_A$, $\pi_C = 0.05 + 0.425\pi_A + 0.85\pi_B$; solving, $\pi = (\frac{686}{1769}, \frac{380}{1769}, \frac{703}{1769}) = (0.388, 0.215, 0.397)$: $C > A > B$. A's one in-link is the only link of the top page C, so it carries all of C's weight; B's in-link is one of two from A and carries half of A's. (c) $\pi \mathbf{H} = \pi$: $\pi_A = \pi_C$, $\pi_B = \frac{1}{2}\pi_A$, $\pi_C = \frac{1}{2}\pi_A + \pi_B$: $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$. (d) Without damping, D is absorbing: once the surfer arrives (which happens, since C is visited again and again) it never leaves, and the only stationary distribution is $\pi = (0, 0, 0, 1)$. With $d = 0.85$, solving $\pi \mathbf{P} = \pi$ gives $\pi = (0.101, 0.080, 0.148, 0.671)$: D still holds two thirds of the mass, because each visit lasts a geometric number of steps with mean $1/0.15 = 6.7$ and every jump lands on D a quarter of the time. Damping makes the problem finite but does not neutralise a self-linking trap; real search engines treat such pages separately.

C17. A chain on $S = \{1, \dots, 6\}$ with

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0.2 & 0 & 0.4 & 0 & 0.4 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0.5 & 0.5 \end{pmatrix}.$$

(a) Find the communicating classes, say which are closed, give the period of each class, and classify each state as recurrent or transient. (b) Compute the probability that the chain started at 3, and at 4, ends up in the class $\{1, 2\}$. (c) Compute the expected time to leave $\{3, 4\}$ from 3 and from 4. (d) Find all stationary distributions. (e) Started at 3, does the distribution of X_n converge? Describe its behaviour for large n .

Solution. (a) $\{1, 2\}$: closed, $1 \rightarrow 2 \rightarrow 1$ only, period 2. $\{5, 6\}$: closed, $p(6, 6) > 0$, period 1. $\{3\}$: $3 \rightarrow 1$ and $3 \rightarrow 5$ leave it, so not closed; $p(3, 3) > 0$, period 1. $\{4\}$: $4 \rightarrow 3$ leaves it, not closed, period 1. States 1, 2, 5, 6 recurrent (finite closed classes); 3, 4 transient. (b) $h_3 = 0.2 \cdot 1 + 0.4h_3 + 0.4 \cdot 0$, so $h_3 = \frac{1}{3}$; $h_4 = 0.5h_3 + 0.5h_4$, so $h_4 = h_3 = \frac{1}{3}$. With probability $\frac{2}{3}$ the chain ends in $\{5, 6\}$. (c) $k_3 = 1 + 0.4k_3$: $k_3 = \frac{5}{3}$; $k_4 = 1 + 0.5k_3 + 0.5k_4$: $k_4 = 2 + k_3 = \frac{11}{3}$. (d) On $\{1, 2\}$ the unique stationary distribution is $(\frac{1}{2}, \frac{1}{2})$; on $\{5, 6\}$, $\pi_5 = 0.5\pi_6$ gives $(\frac{1}{3}, \frac{2}{3})$. Transient states carry no mass: if $\pi_3 + \pi_4 > 0$ then, since $\{3, 4\}$ is left for good with positive probability at every step and never re-entered from $\{1, 2\}$ or $\{5, 6\}$, $\pi \mathbf{P}^n$ puts mass $\rightarrow 0$ on $\{3, 4\}$, contradicting $\pi \mathbf{P}^n = \pi$; so $\pi_3 = \pi_4 = 0$. All stationary distributions: $\alpha(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0) + (1 - \alpha)(0, 0, 0, 0, \frac{1}{3}, \frac{2}{3})$, $\alpha \in [0, 1]$. (e) No. The mass $\frac{1}{3}$ that enters $\{1, 2\}$ alternates between the two states forever (period 2), so $\mathbb{P}_3(X_n = 1)$ oscillates; the mass $\frac{2}{3}$ in $\{5, 6\}$ settles to $(\frac{2}{9}, \frac{4}{9})$. Both hypotheses of the convergence theorem fail: the chain is reducible, and one closed class is periodic.

C18. Walks on the cycle $\mathbb{Z}_L = \{0, \dots, L - 1\}$, addition modulo L . (a) $L = 6$, steps ± 1 with probability $\frac{1}{2}$ each. Compute $p^{(n)}(0, 0)$ for $n = 1, \dots, 4$, give the period, and describe the rows of $\mathbf{P}^n$ for large n (is π stationary; do the rows converge to it?). (b) $L = 7$, steps ± 1 . Give the period and justify it with two round-trip lengths. (c) $L = 6$, steps $+1$ or $+2$ with probability $\frac{1}{2}$ each. Is the chain irreducible? Aperiodic? What is π , and do the rows of $\mathbf{P}^n$ converge? (d) $L = 6$, steps $+2$ or $+4$ with probability $\frac{1}{2}$ each. Find the communicating classes and the period within each. (e) $L = 4$, step $+1$ with probability one. Find π , the period, and say which of the two long-run theorems applies.

Solution. (a) $p^{(n)}(0, 0) = 0, \frac{1}{2}, 0, \frac{3}{8}$ ($n = 4$: four steps with two of each sign, $\binom{4}{2}/16$; a net displacement of ± 6 is impossible in four steps): every return takes an even number of steps, period 2. The uniform $\pi = \frac{1}{6}(1, \dots, 1)$ is stationary (each column receives $\frac{1}{2}$ from two states), but the rows of $\mathbf{P}^n$ do not converge to it: from 0 the walk is on $\{0, 2, 4\}$ at even times and on $\{1, 3, 5\}$ at odd times, and row 0 of $\mathbf{P}^n$ alternates between (approximately) $(\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0)$ and $(0, \frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3})$. (b) Round trips of length 2 ($+1, -1$) and 7 (seven steps $+1$): $\gcd(2, 7) = 1$, period 1; here the rows of $\mathbf{P}^n$ do converge to the uniform π . (c) Irreducible: $+1$ steps reach every state. Round trips of length 3 ($+2, +2, +2$) and 4 ($+1, +1, +2, +2$): aperiodic. Every column of $\mathbf{P}$ receives $\frac{1}{2}$ from two states, so the uniform $\pi = \frac{1}{6}(1, \dots, 1)$ is stationary, and by the convergence theorem every row of $\mathbf{P}^n$ tends to it. (d) From 0 only 0, 2, 4 are reachable: two closed classes, $\{0, 2, 4\}$ and $\{1, 3, 5\}$; the chain is reducible. Within a class, round trips of length 2 ($+2, +4$) and 3 ($+2, +2, +2$): period 1. (e) Deterministic rotation: π uniform (the only solution of $\pi_{i+1} = \pi_i$), period 4. The

ergodic theorem applies (irreducible): each state is visited exactly a quarter of the time. The convergence theorem does not: $\mathbb{P}_0(X_n = 0)$ is 1 or 0 according to $n \bmod 4$.

Hitting probabilities and times

Lecture battery 10

- C19. Tennis at deuce. From deuce the server wins each point with probability p , independently. States: D (deuce), I (advantage server), O (advantage receiver), W (game won), L (game lost). From D the next state is I or O; from I it is W or D; from O it is D or L. (a) Write the transition matrix. (b) Write the hitting system for $h_i = \mathbb{P}_i(\text{reach W})$ and solve it; show $h_D = \frac{p^2}{p^2 + q^2}$. (c) Write the system for $k_i = \mathbb{E}_i[\text{points until the game ends}]$ and show $k_D = \frac{2}{1 - 2pq}$. (d) Evaluate h_D and k_D for $p = 0.5, 0.55, 0.6, 0.7$. What does a 55/45 edge per point become per game? (e) What is the head-on computation of h_D , and why does the first-step system avoid it?

Solution. (a) Order D, I, O, W, L: $\mathbf{P} = \begin{pmatrix} 0 & p & q & 0 & 0 \\ q & 0 & 0 & p & 0 \\ p & 0 & 0 & 0 & q \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. (b) $h_W = 1$, $h_L = 0$, $h_D = ph_I +$

qh_O , $h_I = p + qh_D$, $h_O = ph_D$. Substituting, $h_D = p^2 + pqh_D + pqh_D$, so $h_D(1 - 2pq) = p^2$ and, as $1 - 2pq = p^2 + q^2$, $h_D = \frac{p^2}{p^2 + q^2}$. For $p = 0.6$: $h_D = \frac{9}{13}$, $h_I = \frac{57}{65}$, $h_O = \frac{27}{65}$. (c) $k_W = k_L = 0$, $k_D = 1 + pk_I + qk_O$, $k_I = 1 + qk_D$, $k_O = 1 + pk_D$. Then $k_D = 1 + p + q + 2pqk_D = 2 + 2pqk_D$, so $k_D = \frac{2}{1 - 2pq}$; for $p = 0.6$, $k_D = \frac{50}{13} = 3.85$ points. (d) $h_D = 0.5, 0.599, 0.692, 0.845$ and $k_D = 4, 3.96, 3.85, 3.45$. A 55/45 point becomes a 60/40 game from deuce; the edge compounds because the game only ends when one player is two points ahead. (e) The game ends after $2m$ more points with probability $(2pq)^{m-1}(p^2 + q^2)$, and the server wins with probability $\sum_{m \geq 1} (2pq)^{m-1}p^2$: an infinite series. The first-step system uses the restart at deuce to write h_D in terms of itself.

- C20. The taxi chain of C11 again. (a) Compute the expected number of trips until the next trip starting in C, from A and from B. (b) Compute the mean return time to C by first-step analysis and compare with $1/\pi_C$. (c) Make C absorbing and compute the fundamental matrix $\mathbf{N}$; recover (a) from $\mathbf{N}\mathbf{1}$ and read off the expected number of trips starting in B before the next C-start, when the current trip starts in A.

Solution. (a) $k_C = 0$; $k_A = 1 + 0.5k_A + 0.3k_B$, $k_B = 1 + 0.2k_A + 0.6k_B$. The second gives $0.4k_B = 1 + 0.2k_A$, the first $0.5k_A = 1 + 0.3k_B$; solving, $k_A = k_B = 5$ trips. (b) $\mathbb{E}_C[T_C] = 1 + 0.3k_A + 0.3k_B + 0.4 \cdot 0 = 1 + 1.5 + 1.5 = 4 = 1/\pi_C$. (c) $\mathbf{Q} = \begin{pmatrix} 0.5 & 0.3 \\ 0.2 & 0.6 \end{pmatrix}$, $\mathbf{I} - \mathbf{Q} = \begin{pmatrix} 0.5 & -0.3 \\ -0.2 & 0.4 \end{pmatrix}$, determinant 0.14, $\mathbf{N} = \frac{1}{0.14} \begin{pmatrix} 0.4 & 0.3 \\ 0.2 & 0.5 \end{pmatrix} = \begin{pmatrix} \frac{20}{7} & \frac{15}{7} \\ \frac{10}{7} & \frac{25}{7} \end{pmatrix}$; $\mathbf{N}\mathbf{1} = (5, 5)$. $\mathbf{N}_{AB} = \frac{15}{7} = 2.14$ trips start in B, on average, before the next C-start.

Convergence, the ergodic theorem, absorbing chains

Lecture batteries 11–12

- C21. Long-run behaviour of the taxi chain of C11 and C15. (a) The eigenvalues of $\mathbf{P}$ are 1, 0.3 and 0.2 (check the trace and the determinant). How fast do the rows of $\mathbf{P}^n$ approach π ? The maximal deviations are 0.179, 0.051, 0.015 at $n = 1, 2, 3$; compare with 0.3^n . (b) State what fraction of trips start in each zone in the long run, and which theorem says so. (c)

What fraction of trips go from A to B in the long run? (d) Trips starting in A, B, C earn £10, £8, £12. What is the long-run average fare per trip? (e) For the rotation on $\mathbb{Z}_4$ of C18(e), the long-run fraction of time at each state is $\frac{1}{4}$ but $\mathbb{P}_0(X_n = 0)$ does not converge. Which hypothesis of the convergence theorem fails, and why does the ergodic theorem not need it?

Solution. (a) Trace $0.5 + 0.6 + 0.4 = 1.5 = 1 + 0.3 + 0.2$; determinant $0.5(0.24 - 0.06) - 0.3(0.08 - 0.06) + 0.2(0.06 - 0.18) = 0.06 = 1 \cdot 0.3 \cdot 0.2$. The error shrinks by a factor of about 0.3 per step: 0.179, 0.051, 0.015 have successive ratios 0.29, 0.30. (b) $\pi = (\frac{9}{28}, \frac{3}{7}, \frac{1}{4})$: 32.1%, 42.9%, 25% of trips, with probability one along the actual sequence of trips (ergodic theorem, irreducible finite chain). (c) Ergodic theorem for the pair chain (X_n, X_{n+1}) : $\pi_{AP}(A, B) = \frac{9}{28} \cdot 0.3 = \frac{27}{280} = 9.6\%$. (d) $\sum_i \pi_i f(i) = \frac{9}{28} \cdot 10 + \frac{3}{7} \cdot 8 + \frac{1}{4} \cdot 12 = \frac{135}{14} = £9.64$ per trip. (e) Aperiodicity fails (period 4). The ergodic theorem counts visits along a path; the fraction of time at 0 is exactly $\frac{1}{4}$ whether or not the visits come at regular intervals. The convergence theorem asks where the chain is at one fixed late time, and a periodic chain still remembers the starting time.

C22. The fundamental matrix. (a) Fair ruin chain on $\{0, \dots, 5\}$, transient states 1, 2, 3, 4. Write $\mathbf{Q}$ and $\mathbf{R}$, compute $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$, and check $\mathbf{N}\mathbf{1}$ against $i(5 - i)$ and the target column of $\mathbf{NR}$ against $i/5$. Starting from £1, how many times on average is the fortune £3 before the game ends, and how many times £1 (counting the start)? (b) Tennis, $p = 0.6$ (C19). Compute $\mathbf{N}$ for the transient states D, I, O and confirm $h_D = \frac{9}{13}$ and $k_D = \frac{50}{13}$ from $\mathbf{NR}$ and $\mathbf{N}\mathbf{1}$. Interpret $\mathbf{N}_{DD}$ and check it against the visits criterion $\mathbf{N}_{ii} = 1/(1 - f_i)$ with $f_D = \mathbb{P}_D(\text{return to D})$.

Solution. (a) $\mathbf{Q}$ is 4×4 with $\frac{1}{2}$ on the two off-diagonals and zeros elsewhere; $\mathbf{R}$ has $\frac{1}{2}$ in row

1, column 0 and in row 4, column 5. $\mathbf{N} = \frac{1}{5} \begin{pmatrix} 8 & 6 & 4 & 2 \\ 6 & 12 & 8 & 4 \\ 4 & 8 & 12 & 6 \\ 2 & 4 & 6 & 8 \end{pmatrix}$ (check: $(\mathbf{I} - \mathbf{Q})\mathbf{N} = \mathbf{I}$ row by

row). $\mathbf{N}\mathbf{1} = (4, 6, 6, 4) = i(5 - i)$; $\mathbf{NR}$ has target column $(\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}) = i/5$. From £1: $\mathbf{N}_{13} =$

$\frac{4}{5}$ visits to £3, $\mathbf{N}_{11} = \frac{8}{5}$ visits to £1. (b) $\mathbf{Q} = \begin{pmatrix} 0 & 0.6 & 0.4 \\ 0.4 & 0 & 0 \\ 0.6 & 0 & 0 \end{pmatrix}$, $\mathbf{R} = \begin{pmatrix} 0 & 0 \\ 0.6 & 0 \\ 0 & 0.4 \end{pmatrix}$ (columns

W, L), $\mathbf{N} = \frac{1}{13} \begin{pmatrix} 25 & 15 & 10 \\ 10 & 19 & 4 \\ 15 & 9 & 19 \end{pmatrix}$; $\mathbf{N}\mathbf{1} = (\frac{50}{13}, \frac{33}{13}, \frac{43}{13})$ and $\mathbf{NR}$ has first column $(\frac{9}{13}, \frac{57}{65}, \frac{27}{65})$, as

in C19. $\mathbf{N}_{DD} = \frac{25}{13} = 1.92$ is the expected number of deuces in the game, the first included; $f_D = 2pq = 0.48$ (one point each way), and $1/(1 - 0.48) = \frac{25}{13}$.